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Robust implementation strategy for k-omega SST turbulence model with immersed boundary methods

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ABSTRACT

For Reynolds-averaged Navier–Stokes (RANS) simulations on Cartesian grids with immersed boundaries, a reformulated $k-\omega$ shear-stress transport (SST) model and its wall modeling are proposed. To avoid divergence of ω at the wall, the proposed methodology adopts $\tau \equiv \omega^{-1}$ as the alternative variable, whose transport equation is derived analytically from the ω equation. Furthermore, a wall damping function is introduced to extend the log-layer solutions of k and τ to the wall. The resulting turbulence transport variables have constant or linear profiles in the near-wall region, which are sufficiently resolved by the grid used in wall-modeled simulations. Additionally, an analytical algebraic wall model is developed from the near-wall similarity solution of the turbulence model variables. On wall-resolving structured grids, the developed turbulence model exhibits improved grid-convergence behavior compared to the original $k-\omega$ SST model. Furthermore, for non-body-conforming Cartesian grids, the proposed models are combined with an immersed boundary method using the partial-slip velocity boundary condition and the derived wall model. In validation cases, including the flat-plate turbulent boundary layer, the flow around the NACA0012 airfoil, and the flow past a backward-facing step, the simulation results with the present models show good agreement with the reference results with the $k-\omega$ SST model on body-conforming grids. Moreover, the present model is applied to the flow simulation around a civil aircraft model, showing reasonable prediction of the surface pressure and aerodynamic coefficients.

1. Introduction

Hierarchical Cartesian grids have become increasingly popular in computational fluid dynamics simulations because they can easily handle complex geometries. Since hierarchical Cartesian grids can be generated automatically using tree-based algorithms, they are also suitable for shape optimization problems that require simulating a wide range of geometries. Furthermore, the orthogonality and uniformity of the grids result in the stability and accuracy of the flow simulations. Despite such advantages, a major drawback of (hierarchical) Cartesian grids is the representation of the wall boundary. Without any treatments, the wall is represented as a staircase, significantly reducing the predictability of the near-wall aerodynamics. To recover a smooth wall representation on Cartesian grids, various techniques, collectively known as immersed boundary (IB) methods, have long been studied (see review papers [1,2]).

On Cartesian grids, the cell aspect ratio is generally fixed to unity. Therefore, even using the IB methods, it is impractical to resolve the viscous sublayer of turbulent boundary layers. To overcome this difficulty, coupling with wall models is required for simulating

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high-Reynolds-number flows at a realistic computational cost. Historically, the combination of IB methods and wall models has been tested by several researchers for Reynolds-averaged Navier–Stokes (RANS) simulations (e.g., Kalitzin and Iaccarino [3]) or large-eddy simulations (e.g., Tessicini et al. [4], Roman et al. [5]). However, since the IB methods generally do not satisfy strict mass and momentum conservation (except for the cut-cell methods [6,7]), a naïve implementation of the wall model can cause significant numerical oscillation and inaccuracy when the grid is not conforming to the wall boundary [8].

The inaccuracy on non-body-conforming grids stems from the low grid resolution in the near-wall region. With typical wall modeling, only a few grid points exist between the wall and the forcing point in the log layer. Therefore, the velocity profile assumed by the wall law is not correctly resolved by the grid. On non-body-conforming grids, this under-resolution causes errors in conservation laws, leading to non-physical momentum in- or out-flux at the wall. This excessive flux leads to an inaccurate prediction of boundary-layer development. To address the numerical challenges in resolving the near-wall velocity profiles, one approach is to directly impose the wall-model solutions on the fluid cells near the wall [9,10]. The accuracy of this method depends on the number of cell layers from the wall where the solutions are imposed; for example, Constant et al. [9] recommended four layers for the imposition. Another approach [8,11–14] employs linearization of the velocity profiles in the near-wall region. The linearized velocity profile can be easily reproduced on the grid, reducing numerical oscillations and improving numerical accuracy. Since the velocity is linearized, the eddy viscosity should be modified to maintain the shear-stress balance in the near-wall region, as discussed in the RANS [8] and LES [12] contexts.

In addition to the velocity profile, the boundary conditions for the turbulent model equations can also be problematic. This issue is not the case for the Spalart–Allmaras (SA) turbulence model [15], since its transport variable $\tilde{\nu}$ has a linear solution in the near-wall region [16,17] (i.e., the solution is proportional to the wall distance d). However, some other turbulence models have non-linear variable profiles in the near-wall region. For example, the specific dissipation ω of the $k-\omega$ models [18,19] and the Reynolds stress model [20] is proportional to d^{-2} and diverges to infinity at the wall [16]. Also, both the kinetic energy k and ω exhibit different asymptotic behaviors between the viscous sublayer and log layer. Therefore, it is difficult to accurately reproduce these different behaviors on the relatively coarse grids used in wall-modeled simulations. To avoid the steep distribution of ω, Capizzano [11] employs $g = 1/\sqrt{\omega}$ as an alternative transport variable to ω. Since g is zero at the wall, this variable transformation remedies the numerical difficulty on coarse grids. More recently, Eisfeld et al. [20] introduced an alternative transport variable $\tilde{\omega} = \log(\omega)$ to relax the steep variable gradient in the near-wall region. Although these approaches reduce numerical difficulties in resolving variable profiles to some extent, they do not adequately account for the variable behavior in the viscous sublayer and log layer.

The objective of this paper is to establish RANS simulation methodologies using the $k-\omega$ shear-stress transport (SST) turbulence model [21], particularly for Cartesian grids with immersed boundaries. Although SA-based models have been widely used with immersed boundary methods [8,13,14], two-equation turbulence models, such as the $k-\omega$ SST model, are also desirable for several reasons. First, as a cross-validation, a flow solver should be equipped with different turbulence models. Furthermore, since the two-equation turbulence model incorporates two transport variables, a wider variety of length or time scales can be introduced. The variety of scales is advantageous for developing RANS or hybrid RANS-LES turbulence models (e.g., the delayed detached-eddy simulations [22,23] or zonal detached-eddy simulations [24,25]). Moreover, since these models incorporate the nature of the turbulent kinetic energy (TKE) equation, extended modeling, such as an explicit transfer from modeled to resolved turbulence using linear forcing [26,27], is enabled. To achieve accurate flow simulations on non-body-conforming Cartesian grids, we introduce a variable transformation and a viscous damping function into the transport equations of the $k-\omega$ SST model. Employing these treatments, the solutions of the turbulent variables can be expressed by a single formula throughout the viscous sublayer and log layer. Since these solutions are linear or constant with wall distance, they can be easily reproduced even on coarse grids used in wall-modeled flow simulations.

The structure of this paper is as follows. First, the turbulence modeling, including the variable transformation and wall damping, is described in Section 2. Then, the coupling with the IB method is explained in Section 3. Next, Section 4 presents the validation studies, and Section 5 concludes the paper.

2. Turbulence models

2.1. Baseline turbulence model

The following definitions are based on the Turbulence Modeling Resource (TMR) [28]. The $k-\omega$ SST model [21] solves the transport equations of the kinetic energy k and specific dissipation ω. The model employs the eddy-viscosity approximation for the Reynolds stress tensor τ_{ij}:

$$\tau_{ij} = 2\mu_t\left(S_{ij} - \frac{1}{3}S_{mm}\delta_{ij}\right) - \frac{2}{3}\rho k\delta_{ij}, \tag{1}$$

where μ_t is the eddy viscosity, $S_{ij} = (\partial u_i/\partial x_j + \partial u_j/\partial x_i)/2$ is the strain tensor, and ρ is the density, and (i, j, m) are dimensional indices. In the $k-\omega$ SST model, μ_t is calculated as

$$\mu_t = \frac{\rho k}{\max(\omega, F_2\Omega/a_1)}, \tag{2}$$

where

$$F_2 = \tanh(\mathrm{arg}_2^2),\ \mathrm{arg}_2 = \max\left(2\frac{\sqrt{k}}{\beta^*\omega d}, \frac{500\nu}{d^2\omega}\right), \tag{3}$$

and $a_1 = 0.31$, $\beta^* = 0.09$. Here, $\Omega \equiv \sqrt{(\partial u_m/\partial x_n - \partial u_n/\partial x_m)^2/2}$ is the vorticity magnitude, d is the distance to the nearest wall, u_i is the flow velocity, x_i is the coordinate, and n is the dimensional index. The transport equations of k and ω in the compressible form are

$$\frac{\partial \rho k}{\partial t} + \frac{\partial \rho u_j k}{\partial x_j} = P_k - \beta^* \rho k\omega + \frac{\partial}{\partial x_j}\left[(\mu + \sigma_k \mu_t)\frac{\partial k}{\partial x_j}\right], \tag{4}$$

$$\frac{\partial \rho\omega}{\partial t} + \frac{\partial \rho u_j \omega}{\partial x_j} = \frac{\gamma P_k}{\nu_t} - \beta\rho\omega^2 + \frac{\partial}{\partial x_j}\left[(\mu + \sigma_\omega \mu_t)\frac{\partial \omega}{\partial x_j}\right] + (1 - F_1)\mathrm{CD}_{k\omega}. \tag{5}$$

The last term of Eq. (5) is the cross-diffusion between k and ω:

$$\mathrm{CD}_{k\omega} = \frac{2\rho\sigma_{\omega 2}}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}, \tag{6}$$

and

$$F_1 = \tanh(\mathrm{arg}_1{}^4),\ \mathrm{arg}_1 = \min\left[\max\left(\frac{\sqrt{k}}{\beta^*\omega d}, \frac{500\nu}{d^2\omega}\right), \frac{4\rho\sigma_{\omega 2}k}{\max(\mathrm{CD_{k\omega}}, 10^{-20})d^2}\right]. \tag{7}$$

F_1 is unity in the viscous sublayer and log layer, and approaches zero outside the boundary layer. The model coefficients σ_k, σ_ω, β, and γ are blended through F_1 as

$$\phi = F_1\phi_1 + (1 - F_1)\phi_2, \tag{8}$$

where

$$\sigma_{k1} = 0.85,\ \sigma_{k2} = 1.0,\ \sigma_{\omega 1} = 0.5,\ \sigma_{\omega 2} = 0.856,\ \beta_1 = 0.075,\ \beta_2 = 0.0828, \tag{9}$$

and

$$\gamma_1 = \frac{\beta_1}{\beta^*} - \frac{\sigma_{\omega 1}\kappa^2}{\sqrt{\beta^*}},\ \gamma_2 = \frac{\beta_2}{\beta^*} - \frac{\sigma_{\omega 2}\kappa^2}{\sqrt{\beta^*}},\ \kappa = 0.41. \tag{10}$$

The production P_k is originally defined as $\tau_{ij}\partial u_i/\partial x_j$, whereas slightly different forms have been proposed in previous studies [18,21]. Most commonly, the production is modeled as

$$P_k = \min\left(\mu_t S^2, 20\beta^*\rho k\omega\right), \tag{11}$$

where $S = \sqrt{(\partial u_m/\partial x_n + \partial u_n/\partial x_m)^2/2}$ is the strain magnitude. Also, the "-V" variant [18] uses Ω^2 instead of S^2.

One of the challenges in the $k-\omega$ SST model is the near-wall behavior of the turbulent variables. As summarized by Kalitzin et al. [16], the solutions of the turbulent variables in the viscous sublayer are

$$k_{\mathrm{vis}} = c_k u_\tau^{5.23}\left(\frac{d}{\nu}\right)^{3.23},\ \omega_{\mathrm{vis}} = \frac{6\nu}{\beta_1 d^2}, \tag{12}$$

and those in the log layer are

$$k_{\log} = \frac{u_\tau^2}{\sqrt{\beta^*}},\ \omega_{\log} = \frac{u_\tau}{\kappa\sqrt{\beta^*}d}. \tag{13}$$

The coefficient c_k in Eq. (12) is approximately 0.003 in the flat-plate turbulent boundary-layer (FPTBL) in Section 4.1, although no analytical formula is provided for this coefficient. Since typical computational grids for wall-modeled simulations do not resolve the viscous sublayer, these near-wall variable distributions may not be accurately reproduced on the grid. In particular, the divergence of ω at the wall is problematic.

2.2. Variable transformation

As the transport variable that substitutes for ω, we introduce $\tau \equiv \omega^{-1}$, which has the dimension of time. The advantage of τ is that it is proportional to the wall distance in the log layer and becomes zero at the wall. The transport equation for τ is derived from Eq. (5) by applying the relationship $d\tau = -\tau^2 d\omega$:

$$\begin{aligned}\frac{\partial \rho\tau}{\partial t} + \frac{\partial \rho u_j \tau}{\partial x_j} &= -\tau^2\left[(\text{RHS of } \omega \text{ Eq.})\right] \\ &= -\gamma\rho S^2\tau^2 + \beta\rho + \tau^2\frac{\partial}{\partial x_j}\left[(\mu + \sigma_\omega\mu_t)\frac{1}{\tau^2}\frac{\partial \tau}{\partial x_j}\right] + (1 - F_1)\mathrm{CD}_{k\tau},\end{aligned} \tag{14}$$

where

$$\mathrm{CD}_{k\tau} = 2\rho\sigma_{\omega 2}\tau\frac{\partial k}{\partial x_j}\frac{\partial \tau}{\partial x_j}. \tag{15}$$

The auxiliary functions of the $k-\omega$ SST model can be calculated through the relationships $\omega = \tau^{-1}$ and $\partial\omega/\partial x_j = -\tau^{-2}\partial\tau/\partial x_j$. Note that a similar variable transformation is adopted in the previous research [29], which derived the τ equation from the k and ε (dissipation) equations.

2.3. Extending the log-layer solution to the wall

The transport variable τ is proportional to d in the log layer but to d^2 in the viscous sublayer. This behavior is problematic when applying the wall model because the smooth transition between the viscous sublayer and log-layer solutions is not guaranteed. Therefore, we introduce a viscous damping function to allow the log-layer solutions of k and τ to extend to the wall.

The solutions of the $k-\tau$ model equations in the log layer are

$$k_{\log} = \frac{u_\tau^2}{\sqrt{\beta^*}}, \ \tau_{\log} = \frac{\kappa\sqrt{\beta^*}d}{u_\tau}. \tag{16}$$

With these solutions, the production and dissipation terms of the k equation become equal to $\rho u_\tau^3/(\kappa d)$, whereas the diffusion term is zero. To retain these solutions up to the wall, we introduce surrogate variables $\tilde{k}$ and $\tilde{\tau}$, which satisfy

$$\mu_t = \rho\tilde{k}\tilde{\tau} f_{v1}, \tag{17}$$

except when the eddy viscosity limiter of the SST model (the maximum function in Eq. (2)) is active. The function f_{v1} is a wall-damping function that is similar to the SA turbulence model [15]. The concrete form of f_{v1} for the present model is considered in Section 2.5. With this damping function, we assume that the following near-wall solutions are satisfied throughout the viscous sublayer and log-layer:

$$\tilde{k}_{\mathrm{nw}} = \frac{u_\tau^2}{\sqrt{\beta^*}}, \ \tilde{\tau}_{\mathrm{nw}} = \frac{\kappa\sqrt{\beta^*}d}{u_\tau}. \tag{18}$$

To retain the log-layer solution in the viscous sublayer, the term including the velocity gradient must be modified. Here, S^2 in the production terms is replaced by

$$\Theta = \max\left[S^2 + \left(\frac{\sqrt{\beta^*}}{\tilde{\tau}}\right)^2 f_{v2}F_1,\ (0.3S)^2\right],\ f_{v2} = 1 - \left(\frac{\chi}{1+\chi f_{v1}}\right)^2, \tag{19}$$

where $\chi = \tilde{k}\tilde{\tau}/\nu$. The function form of f_{t2} is borrowed from $\tilde{S}$ of the SA turbulence model, except for the exponent 2. Note that the lower limit $(0.3S)^2$ is added to avoid negative production because, although very rarely, the term containing f_{v2} becomes smaller than $-S^2$. In both the viscous sublayer and log layer, Θ defined by Eq. (19) equals to $(u_\tau/(\kappa d))^2$. This is confirmed as follows. Here, we assume that F_1 is unity and the lower limit $(0.3S)^2$ can be ignored, which is true at a typical location within the near-wall region. Also, from the thin-layer approximation of the momentum equation, the following equation is satisfied:

$$(\mu + \mu_t)S = \rho u_\tau^2, \tag{20}$$

i.e.,

$$S = \frac{u_\tau^2}{\nu(1+\chi f_{v1})}. \tag{21}$$

With Eq. (18), $\chi \equiv \mu_t/(\rho\nu f_{v1})$ is equal to $\kappa u_\tau d/\nu$, and hence,

$$S = \frac{\kappa u_\tau d/\nu}{1+\chi f_{v1}}\frac{u_\tau}{\kappa d} = \frac{\chi}{1+\chi f_{v1}}\frac{u_\tau}{\kappa d}. \tag{22}$$

Also, from Eq. (18),

$$\frac{\sqrt{\beta}^*}{\tilde{\tau}} = \frac{u_\tau}{\kappa d}. \tag{23}$$

Thus, substituting Eqs. (22) and (23) into Eq. (19) yields $\Theta = (u_\tau/(\kappa d))^2$.

Furthermore, the production is modeled as $P_{\tilde{k}} = \mu_t\Theta/f_{v1}$ so that both the production and dissipation terms become equal to $\rho u_\tau^3/(\kappa d)$. Although these terms become infinite at the wall, these values are evaluated at the cell center with a finite wall distance. Also, these terms do not include derivatives of quantities that diverge at the wall. For these reasons, numerical instability is unlikely to occur, and indeed it does not occur in the validation problems in Section 4. In the τ equation, S^2 is similarly replaced by Θ, and the molecular diffusion is eliminated in the viscous sublayer by multiplying the molecular viscosity by $1-F_1$. Consequently, with Eq. (18) and $\Theta = (u_\tau/(\kappa d))^2$, each term has the following solution throughout the viscous sublayer and log layer:

$$\begin{aligned}
-\gamma\rho\Theta\tilde{\tau}^2 &= -(\beta_1 - \kappa^2\sigma_{\omega 1}\sqrt{\beta^*})\rho, \\
\beta\rho &= \beta_1\rho, \\
\tilde{\tau}^2\frac{\partial}{\partial x_j}\left[\left((1-F_1)\mu + \sigma_\omega\rho\tilde{k}\tilde{\tau}\right)\frac{1}{\tilde{\tau}^2}\frac{\partial\tilde{\tau}}{\partial x_j}\right] &= -\kappa^2\sigma_{\omega_1}\sqrt{\beta^*}\rho,
\end{aligned} \tag{24}$$

while the cross-diffusion is zero. Since these source terms balance, the governing equation is satisfied in the near-wall region, where the advection term and time derivative are negligible. Note that S^2 in Eq. (19) is replaced with Ω^2 in the "-V" variant, like the $k-\omega$ SST-V model. Also, when the exact production term is used, S^2 is replaced by $(\tau_{ij}\partial u_i/\partial x_j)/\mu_t$.

2.4. Summary of the model equations

The complete governing equations of the $\tilde{k}-\tilde{\tau}$ SST model are as follows:

$$\frac{\partial \rho \tilde{k}}{\partial t}+\frac{\partial \rho u_j \tilde{k}}{\partial x_j}=P_{\tilde{k}}-\beta^* \rho \frac{\tilde{k}}{\tilde{\tau}}+\frac{\partial}{\partial x_j}\left[(\mu+\sigma_k \rho \tilde{k} \tilde{\tau}) \frac{\partial \tilde{k}}{\partial x_j}\right], \tag{25}$$

$$\frac{\partial \rho \tilde{\tau}}{\partial t}+\frac{\partial \rho u_j \tilde{\tau}}{\partial x_j}=-\gamma \rho \tilde{\tau}^2 \Theta+\beta \rho+\tilde{\tau}^2 \frac{\partial}{\partial x_j}\left[\left((1-F_1)\mu+\sigma_\omega \rho \tilde{k} \tilde{\tau}\right) \frac{1}{\tilde{\tau}^2} \frac{\partial \tilde{\tau}}{\partial x_j}\right]+(1-F_1) 2 \sigma_{\omega 2} \rho \tilde{\tau} \frac{\partial \tilde{k}}{\partial x_j} \frac{\partial \tilde{\tau}}{\partial x_j}, \tag{26}$$

where

$$P_{\tilde{k}}=\min \left(\frac{\mu_t}{f_{v 1}} \Theta,\ 20 \beta^* \rho \frac{\tilde{k}}{\tilde{\tau}}\right), \tag{27}$$

and Θ is defined by Eq. (19). The diffusion and cross-diffusion terms in the $\rho\tilde{\tau}$ equation are different from the ordinary conservative form, and their discretizations require special care (see Appendix A). With $\tilde{k}$ and $\tilde{\tau}$, the eddy viscosity is calculated as

$$\mu_t=\rho \tilde{k} \tilde{\tau} f_{v 1} \min \left[1, \frac{a_1}{\max(\tilde{\Omega} \tilde{\tau} F_2, 10^{-12})}\right]. \tag{28}$$

where

$$\tilde{\Omega}=\max \left[\Omega+\frac{\sqrt{\beta^*} F_1}{\tilde{\tau}}\left(1-\frac{\chi}{1+\chi f_{v 1}}\right),\ 0.3 \Omega\right]. \tag{29}$$

The auxiliary functions are calculated as follows:

$$F_1=\tanh(\mathrm{arg}_1{}^4),\ F_2=\tanh(\mathrm{arg}_2{}^2),\ \mathrm{arg}_1=\min(\max(\Gamma_1, \Gamma_2), \Gamma_3),\ \mathrm{arg}_2=\max(2 \Gamma_1, \Gamma_2), \tag{30}$$

where

$$\Gamma_1=\frac{\sqrt{\tilde{k}} \tilde{\tau}}{\beta^* d},\ \Gamma_2=\frac{500 \nu \tilde{\tau}}{d^2},\ \Gamma_3=-\frac{2 \tilde{k} \tilde{\tau}}{d^2} \min \left[\left(\frac{\partial \tilde{k}}{\partial x_j} \frac{\partial \tilde{\tau}}{\partial x_j}\right),-10^{-12} \frac{\nu_{\infty}}{L^2}\right]^{-1}. \tag{31}$$

Here, L is the characteristic length, and ν_∞ is the freestream kinematic viscosity. Since the difference between k and $\tilde{k}$ (and τ and $\tilde{\tau}$) only appears in the viscous sublayer, replacing k and τ with $\tilde{k}$ and $\tilde{\tau}$ essentially does not change these functions' behavior. Note that all the model constants are the same as the $k-\omega$ SST model (Eqs. (8)–(10)).

The boundary conditions are as follows. Since the log-layer solution of k (Eq. (16)) is constant in the wall-normal direction, the Neumann condition $\partial\tilde{k}/\partial d=0$ is imposed at the wall. For $\tilde{\tau}$, the Dirichlet condition $\tilde{\tau}=0$ is employed. Note that the other boundary conditions (e.g., the far-field conditions) can be obtained from the boundary conditions of the $k-\omega$ SST model with conversion $\tilde{k}=k$ and $\tilde{\tau}=\omega^{-1}$.

Furthermore, since the TKE transport equation has a nonzero dissipation rate even in the freestream, TKE gradually decreases between the outer boundary and the wall. This may be problematic when simulating outer flows, where the outer boundary is located far from walls. To avoid the decrease in TKE, we introduce a source term to sustain the freestream values of $\tilde{k}$ and $\tilde{\tau}$ similar to the existing study on the $k-\omega$ SST model [30]. The source terms for $\tilde{k}$ and $\tilde{\tau}$ equations are

$$S_{\tilde{k}}=\beta^* \rho \frac{\tilde{k}_{\infty}}{\tilde{\tau}_{\infty}}(1-F_1),\ S_{\tilde{\tau}}=-\beta \rho \frac{\tilde{k}_{\infty}}{\max(\tilde{k}, \tilde{k}_{\infty})} \frac{\tilde{\tau}}{\tilde{\tau}_{\infty}}(1-F_1), \tag{32}$$

respectively.

2.5. Wall damping and velocity wall functions

The wall damping function f_{v1} is modeled as

$$f_{v 1}=\max \left[\frac{\chi^\alpha}{\chi^\alpha+\psi},\ 1-F_1\right]. \tag{33}$$

Here, $1-F_1$ is introduced to retain the behavior of the original SST model in the outer part of the boundary layer (i.e., $\tilde{k}=k$ and $\tilde{\tau}=\tau$ when $F_1=0$). Similar to the wall function derived from the near-wall solution of the SA turbulence model [17], the corresponding wall function derivative is derived from the thin-layer approximation:

$$\frac{d u^+}{d y^+}=\frac{1}{1+\chi f_{v 1}}, \tag{34}$$

where $u^+=u/u_\tau$, $y^+=du_\tau/\nu_w$ and $u_\tau=\sqrt{\tau_w/\rho_w}$, τ_w is the wall shear stress, and the subscript w represents the quantities at the wall. The wall function is obtained by integrating Eq. (34) from the wall with assuming $\chi\equiv\tilde{k}\tilde{\tau}/\nu=\kappa y^+$. The analytical wall function for the SA turbulence model is derived with $\alpha=3$ and $\psi=(7.1)^3$ in ref. [17].

In Eq. (33), α and ψ are chosen to fit the solution of the $k-\omega$ SST model in the FPTBL case (see Section 4.1). Since α should be a convenient number that enables an analytical integration of Eq. (34), we choose $\alpha=1.5$. Furthermore, the constant $\psi=26.5$

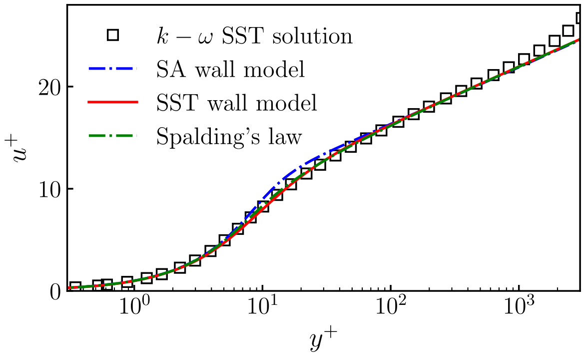


Fig. 1. Comparisons of the wall functions.

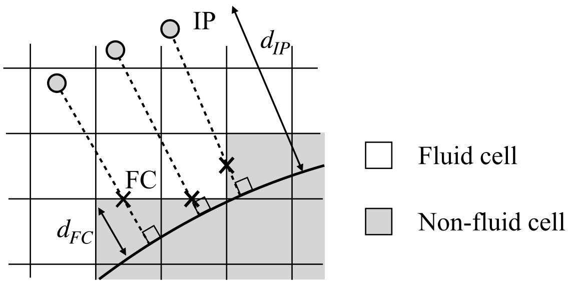


Fig. 2. Schematic of the IB method used in UTCart.

is determined so that C_f in the FPTBL case converges to almost the same value as the original $k - \omega$ SST model. The corresponding wall function is

$$\begin{aligned} u^+(y^+) = & -6.65020 \tan^{-1}\left(-0.539102\sqrt{y^+} + 1.23968\right) \\ & - 4.09533 \tan^{-1}\left(0.331120\sqrt{y^+} + 0.288846\right) \\ & - 0.187302 \log\left(y^+ - 4.59904\sqrt{y^+} + 8.72858\right) \\ & + 4.00770 \log\left(y^+ + 1.74466\sqrt{y^+} + 9.88169\right) \\ & - 2.76275 \log\left(\sqrt{y^+} + 2.85438\right) + 1.20677. \end{aligned} \tag{35}$$

Fig. 1 compares this wall function (herein called the SST wall model) to the wall-resolving solution of the $k - \omega$ SST model in the FPTBL case and other wall functions. Compared to the SA wall model [17], the velocity in the buffer layer ($5 \lesssim y^+ \lesssim 30$) is closer to the $k - \omega$ SST solution, whereas the y-intercept of the logarithmic law is almost the same. Also, the SST wall model resembles Spalding's law [31], whereas it employs the more convenient form $u^+ = f(y^+)$ rather than $y^+ = f(u^+)$ of Spalding's law.

3. IB method with a wall function

Here, the coupling of the $\tilde{k} - \tilde{\tau}$ SST model with the IB method is described. The following description is based on the IB method of UTCart [32], although the methodology may also be applicable to other sharp-interface IB methods. UTCart treats all intersecting cells as non-fluid cells and solves the governing equations only on the fluid cells (see Fig. 2). The wall boundary conditions are imposed at the face center (FC) between the fluid and solid cells as the boundary flux. To determine the wall boundary flux, an image point (IP) is set on the wall-normal line through FC. The distance between the IP and wall, d_{IP}, is a multiple of the local cell size. In this study, we fix $d_{\mathrm{IP}} = 3\Delta x$, following the previous studies using SA-based turbulence models [8,32,33]. The density, velocity, and pressure at the IP are linearly interpolated from the surrounding cells, and the boundary condition at FC is imposed based on the physical relationship. For example, the Neumann condition for the pressure is imposed as $p_{\mathrm{FC}} = p_{\mathrm{IP}}$. For more details of the wall boundary condition, see refs. [8,32].

3.1. Velocity linearization and eddy viscosity augmentation

First, u_τ is calculated using Newton iterations so that u_{IP} and d_{IP} satisfy the wall function. Once u_τ is calculated, the stress tensor at FC is given as $\tau_{ij} = \rho_w u_\tau^2 (t_i n_j + t_j n_i)$. Here, n_i is the i-th component of the wall-normal unit vector, whereas t_i is that of the wall-parallel unit vector, which is obtained by normalizing the wall-parallel velocity vector at IP. Next, the velocity, pressure, and density at the wall are calculated to determine the inviscid flux at FC. As described in the previous studies [8,11], the nonlinear velocity profile in the near-wall region of the turbulent boundary layer cannot be resolved on Cartesian grids in wall-modeled simulations. Using second-order schemes, only linear profiles can be reconstructed within a cell. Therefore, as in the previous study, the velocity is linearized as

$$u_{\mathrm{FC}} = u_{\mathrm{IP}} - \left.\frac{du}{dy}\right|_{\mathrm{IP}} (d_{\mathrm{IP}} - d_{\mathrm{FC}}), \tag{36}$$

where u is the wall-parallel velocity and y is the wall-normal coordinate. Since the slip velocity is nonzero, this boundary condition may be considered a partial-slip boundary condition. In Eq. (36), the velocity gradient is obtained by using the wall function gradient (Eq. (34)) as

$$\left.\frac{du}{dy}\right|_{\mathrm{IP}} = \frac{u_\tau^2}{\nu} \left.\frac{du^+}{dy^+}\right|_{\mathrm{IP}}. \tag{37}$$

Corresponding to the velocity linearization, the eddy viscosity must be augmented to retain the shear-stress balance below the IP [8]. Since the velocity gradient below the IP is constant, the eddy viscosity must also be constant between the IP and the wall. To realize the constant eddy viscosity profile, we modify the wall damping function (Eq. (33)) as

$$f_{v1} = \max\left[\frac{(\chi r_d)^\alpha}{(\chi r_d)^\alpha + \psi},\ 1 - F_1\right] r_d, \tag{38}$$

where $r_d = \max(d_{\mathrm{cutoff}}/d,\ 1.0)$. The cutoff distance d_{cutoff} is the same as the IP distance ($3\Delta x$) at the nearest wall and can be calculated by the advection equation, as described in ref. [32]. Since the equation $\Theta = (u_\tau/(\kappa d))^2$ is satisfied regardless of f_{v1} (see Section 2.3), this modification does not modify the near-wall solutions of the turbulence model equations.

3.2. Boundary conditions for the turbulent transport equations

The boundary conditions for the turbulence model variables are given by the near-wall solution (Eq. (16)) as

$$\tilde{k}_{\mathrm{FC}} = \frac{u_\tau^2}{\sqrt{\beta^*}}, \qquad \tilde{\tau}_{\mathrm{FC}} = \min\left(\frac{\kappa\sqrt{\beta^*} d_{\mathrm{FC}}}{u_\tau},\ \tilde{\tau}_{\mathrm{IP}} \frac{d_{\mathrm{FC}}}{d_{\mathrm{IP}}}\right). \tag{39}$$

Note that the minimum function for $\tilde{\tau}_{\mathrm{FC}}$ in Eq. (39) is introduced to avoid the divergence of $\tilde{\tau}$ when $u_\tau \approx 0$.

4. Validations

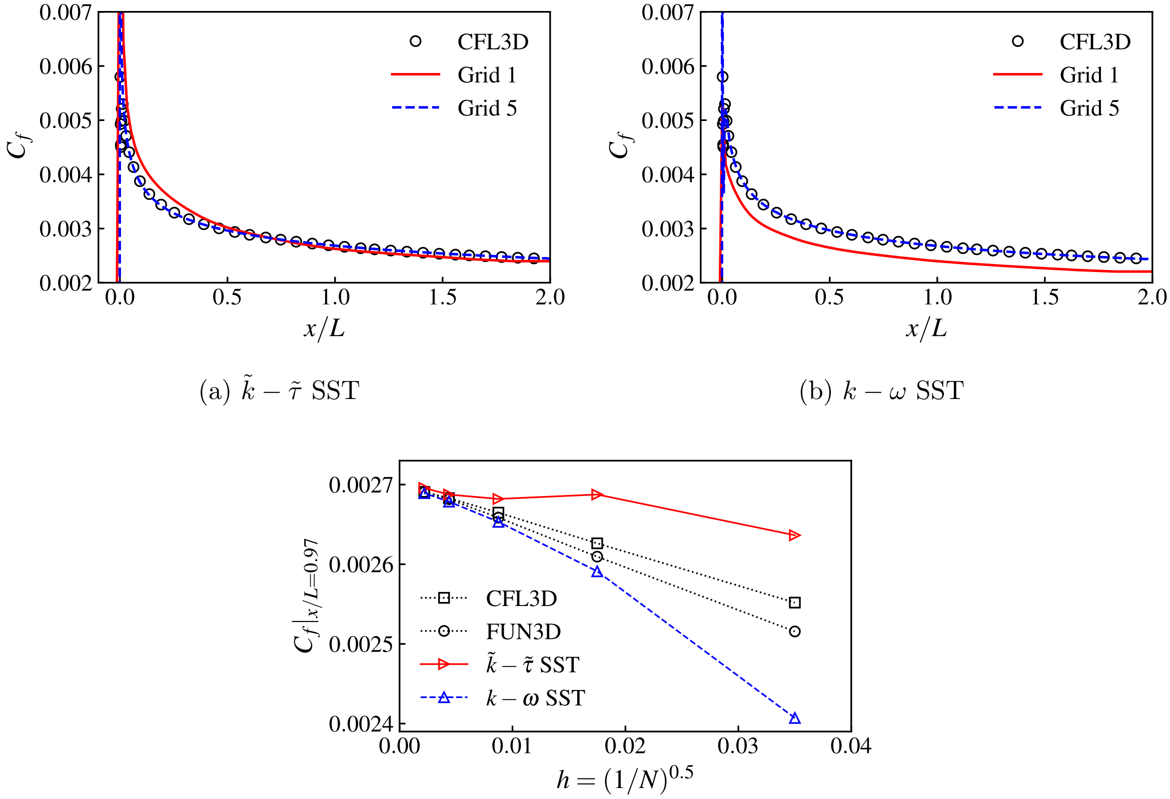
As a preliminary validation, the $\tilde{k} - \tilde{\tau}$ SST model is first implemented in a structured-grid flow solver with a non-slip boundary condition and tested in the FPTBL case. Then, the $\tilde{k} - \tilde{\tau}$ SST model and wall boundary conditions described in Section 3 are implemented in UTCart, and the results are compared to those obtained by structured-grid flow solvers. The primary validation cases in 2D include the FPTBL, the flows around the NACA0012 airfoil, and the flows over a backward-facing step, all of which are defined in TMR [28]. Also, as the extended validation case in 3D, the flows around the National Aeronautics and Space Administration (NASA) Common Research Model are simulated using UTCart with the $\tilde{k} - \tilde{\tau}$ SST model.

In the following test cases, the sustaining terms (Eq. (32)) are applied, except for the internal flow (i.e., the backward-facing step) case. When using the sustaining terms, the farfield values of the turbulent variables are set to $\tilde{k}_\infty = 1.0 \times 10^{-6} u_\infty^2$ and $\tilde{\tau}_\infty = 0.2L/u_\infty$ (i.e., $\mu_{t,\infty}/\mu_\infty = 2 \times 10^{-7} Re_L$), following the recommendation in ref. [30]. In the backward-facing step case, the boundary condition is $\tilde{k}_\infty = 9.0 \times 10^{-9} a_\infty^2$ and $\tilde{\tau}_\infty = 1.0 \times 10^6 \nu_\infty / a_\infty^2$, where ν_∞ is the freestream kinematic eddy viscosity, and a_∞ is the freestream sonic speed. In all the simulations, the TKE term in the viscous stress tensor is omitted. Following the definition in TMR, the employed model should be denoted as "-Vm-sust" or "-Vm" variant, although the notations of these variants are omitted in the following for simplicity.

In both the structured-grid flow solver and UTCart, the inviscid flux is evaluated by the simple low-dissipation advection upstream splitting method (SLAU) [34] with the third-order monotonic upwind central scheme for conservative laws (MUSCL) [35]. No slope limiter is applied to the MUSCL for the mass, momentum, and total energy equations, whereas van Leer's slope limiter [35] is applied to the convection terms of the turbulence model equations to prevent the turbulence variables from being negative at the boundary layer edge. All the computations employ an implicit time-integration method with local time stepping to obtain steady-state solutions.

4.1. FPTBL on body-conforming structured grids

In the FPTBL case, a fully turbulent boundary layer developing over a flat plate of length $2L$ is simulated. The Reynolds number based on u_∞ and L is 5.0×10^6, and the freestream Mach number is 0.2. The results are compared to the reference results obtained



(a) $\tilde{k} - \tilde{\tau}$ SST

(b) $k - \omega$ SST

(c) Convergence of C_f at $x/L = 0.97$

Fig. 3. Skin friction coefficient of the FPTBL case using the structured-grid flow solver.

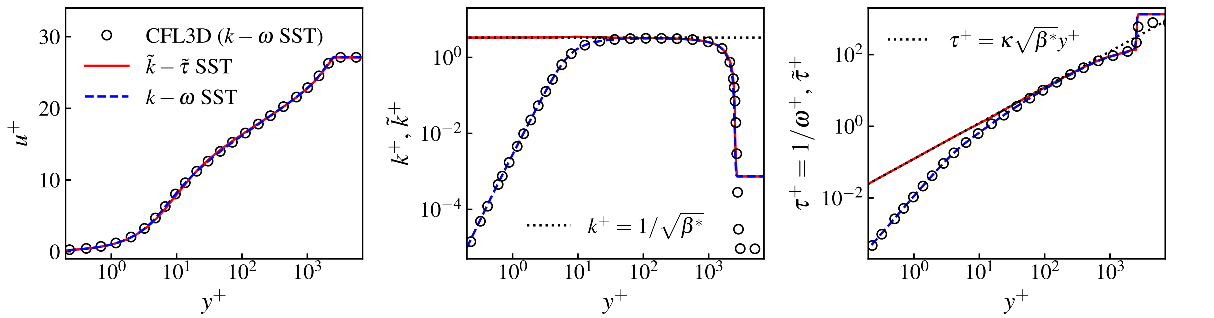
by CFL3D [36] and FUN3D [37], which are provided in TMR. To check the grid-convergence behavior, a series of grids available at TMR is used. The finest grid (Grid 5) has 545×385 grid points, while the coarser grids (Grids 1–4) are successively created by eliminating every two grid points. As a reference, the same case is simulated using the $k - \omega$ SST model on the same flow solver with the freestream boundary conditions as the reference results (i.e., $k_\infty = 9.0 \times 10^{-9} a_\infty^2$ and $\omega_\infty = 10^{-6} a_\infty^2 / \nu_\infty$).

Fig. 3 (a) and (b) compare the distributions of $C_f = \tau_w / (\rho_\infty u_\infty^2 / 2)$ along the plate surface on Grids 1 and 5. On Grid 1, the $k - \omega$ SST model underpredicts C_f by approximately 10%, whereas the $\tilde{k} - \tilde{\tau}$ SST model shows an overall better prediction. On Grid 5, the results from both models are almost identical to the reference solution obtained by CFL3D. Fig. 3 (c) shows the convergence of the C_f at $x/L = 0.97$, where the horizontal axis is the effective cell size, i.e., the square root of the total cell number N. Although both models converge to almost the same value as the reference results ($C_f \approx 0.0027$), the $\tilde{k} - \tilde{\tau}$ SST model exhibits less grid-dependent behavior.

Fig. 4 compares the profiles of the streamwise velocity and turbulent variables in the viscous wall unit on Grid 5. Here, the turbulent variables in the viscous wall unit are defined as $k^+ = k/u_\tau^2$, $\tilde{k}^+ = \tilde{k}/u_\tau^2$, $\omega^+ = \omega\nu/u_\tau^2$, and $\tilde{\tau}^+ = \tilde{\tau} u_\tau^2 / \nu$. The velocity profile obtained by the $\tilde{k} - \tilde{\tau}$ SST model shows excellent agreement with that by the $k - \omega$ SST model. Moreover, except in the viscous sublayer and near the outer edge, the $\tilde{k} - \tilde{\tau}$ SST and $k - \omega$ SST models yield almost identical profiles of the turbulent variables. Note that the differences in the freestream values of k (or $\tilde{k}$) and τ (or $\tilde{\tau}$) between the $k - \omega$ SST model on the present flow solver and the reference solution are due to the sustaining terms, whereas the term has almost no impact on the velocity and turbulent variable profiles inside the boundary layer. The turbulent variable profiles obtained by the $\tilde{k} - \tilde{\tau}$ SST model also reasonably match the theoretical solutions (Eq. (16)) at $y^+ \lesssim 100$, suggesting the model behaves as designed.

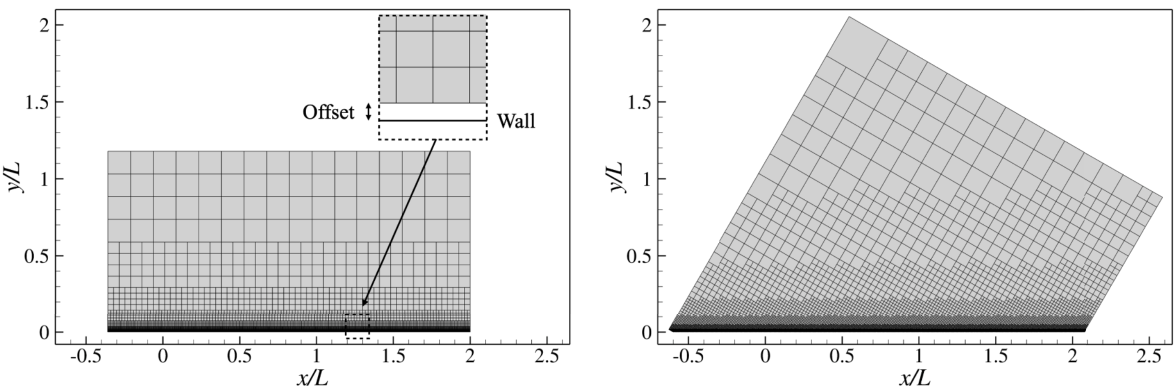
4.2. FPTBL on hierarchical Cartesian grids

The FPTBL case is simulated using UTCart with the $\tilde{k} - \tilde{\tau}$ SST model. For the computational grid, quadtree-based aligned and inclined grids (Fig. 5) are used. On the aligned grid, the offset (see Fig. 5 (a)) is set to $0.5\Delta x$ by default, and the sensitivity to the offset value is separately investigated on Grid 3. On the inclined grid (Fig. 5(b)), the angle between the grid axis and wall is set to 30 degrees so that no streamwise periodicity exists. To check the grid sensitivity, four computational grids with different minimum cell sizes are prepared. Table 1 summarizes the cell sizes and numbers of each computational grid. Since the IP height is $3\Delta x$, the IP height in the viscous wall unit d_{IP}^+ ranges from 80 to 640, which is well within the log layer. Although local grid refinement near the



(a) u^+ (b) k^+ or $\tilde{k}^+$ (c) τ^+ or $\tilde{\tau}^+$

Fig. 4. Profiles of streamwise velocity and turbulent variables at $x/L = 0.97$ on Grid 5.



(a) Aligned grid (b) Inclined grid

Fig. 5. Computational grid for the FPTBL case using UTCart (Grid 1).

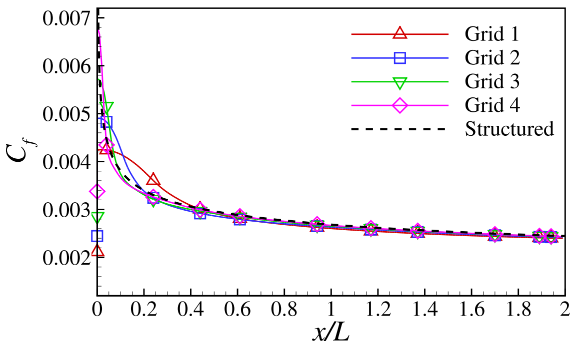
Table 1
Summary of the computational grid for the FPTBL case using UTCart.

Grid	Minimum cell size $\Delta x/L$	Δx^+ at $x/L = 0.97$	Cell number	
			Aligned	Inclined
1	1.15×10^{-3}	210	19,392	28,744
2	5.76×10^{-4}	110	41,872	70,288
3	2.88×10^{-4}	53	108,384	166,090
4	1.44×10^{-4}	26	241,456	383,054

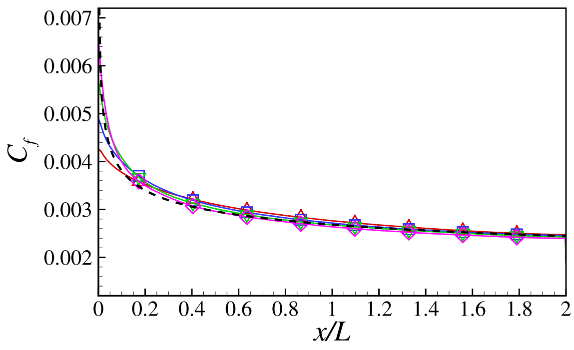
leading edge improves simulation accuracy, a single Δx value is applied across the entire plate surface for the purpose of fundamental validation.

Fig. 6 shows the distributions of C_f along the flat plate. Here, the result on the structured grid (Grid 5 in Section 4.1) is also plotted as a reference solution. Although not shown here, computations using the $k-\omega$ SST model on UTCart encountered numerical instability and failed to converge. On both aligned and inclined grids, the predicted C_f shows good agreement with the reference result on the structured grid in the downstream region $x/L \gtrsim 0.5$, where the error of C_f at $x/L = 0.97$ is smaller than 4% across all the simulated cases. The peak of C_f near the leading edge is not adequately captured on the coarse grids, whereas the prediction is improved on the refined grids. Furthermore, Fig. 7 shows the C_f distributions on the aligned grids with different offsets. The C_f distributions are almost independent of the offset value, except for the minor difference near the leading edge.

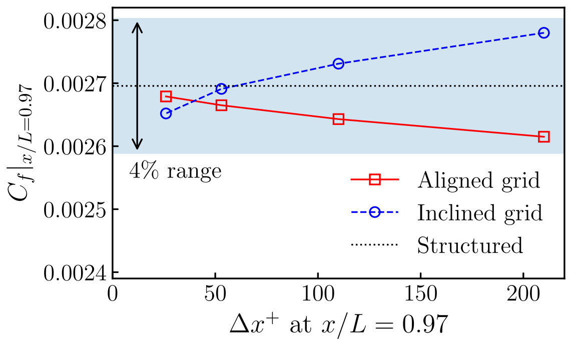
Fig. 8 compares the profiles of the streamwise velocity and turbulence variables at $x/L = 0.97$. All the obtained velocity profiles show good agreement with the reference solution, except in the near-wall region. The deviation of the near-wall velocity profile from the log law shows the effects of the velocity linearization described in Section 3. The turbulence variables also show good agreement with the reference solution, except at the outer edge of the boundary layer. Although the variable distributions near the outer edge tend to be diffusive when the grid is inclined with the flow direction, increasing the grid resolution in this region improves the results.



(a) Aligned grid



(b) Inclined grid



(c) Convergence of C_f at $x/L = 0.97$

Fig. 6. Skin friction coefficient of the FPTBL case using UTCart.

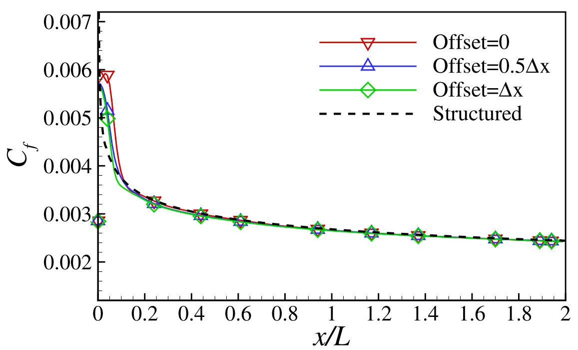
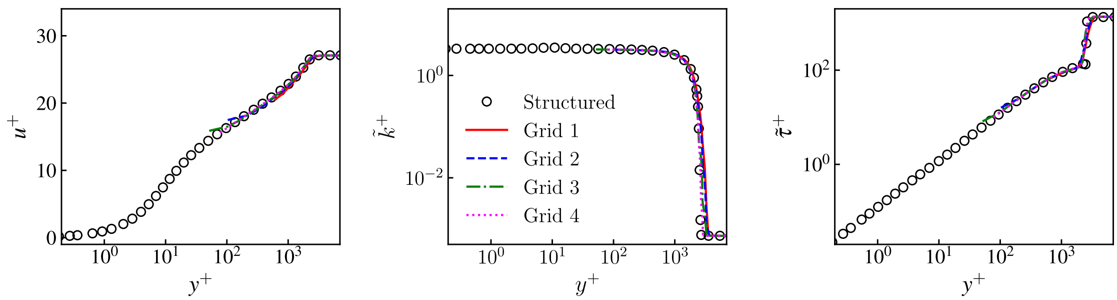


Fig. 7. Sensitivity of the friction coefficient to the offset between the grid and wall (Grid 3).

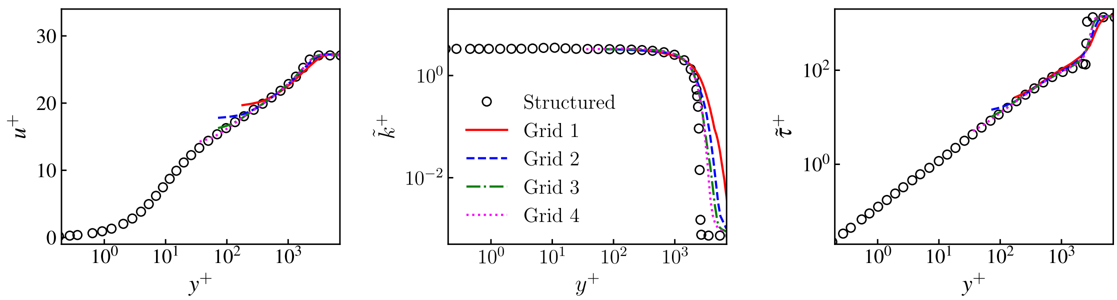
4.3. Flows around the NACA0012 airfoil

The flows around the NACA0012 airfoil are simulated using UTCart with the $\tilde{k} - \tilde{\tau}$ SST model. The freestream Mach number is 0.15, the Reynolds number based on u_∞ and the airfoil chord c is 6.0×10^6, and the angles of attack α are 0, 10, and 15 degrees. Fig. 9 shows the overview of the computational grids. The grid becomes coarser in the farfield, and the outer boundary is located approximately $500c$ from the airfoil. To assess grid sensitivity, four grids with different resolutions are employed, as summarized in Table 2. Furthermore, to obtain reference results, the computation is also conducted using the structured-grid flow solver on the body-fitted grid provided at TMR [28] (897×257 grid points).

Fig. 10 shows the distributions of the chordwise velocity u, the turbulence variables ($\tilde{k}$ and $\tilde{\tau}$), and the eddy viscosity μ_t around the airfoil at $\alpha = 15°$ on Grid 4. In the boundary layer, $\tilde{k}$ and μ_t are significantly larger than the freestream, whereas $\tilde{\tau}$ is smaller than the freestream. Due to the sustaining terms, $\tilde{k}$ and $\tilde{\tau}$ outside the boundary layer are almost identical with the freestream values ($1.0 \times 10^{-6} u_\infty^2$ and $0.2L/u_\infty$, respectively).

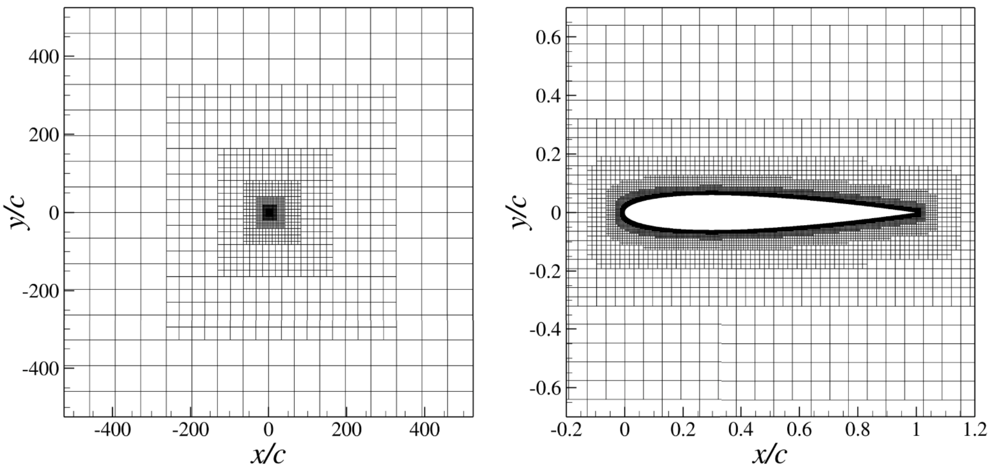


(a) Aligned grid



(b) Inclined grid

Fig. 8. Profile of the streamwise velocity and turbulent variables at $x/L = 0.97$.



(a) Overview (b) Near the airfoil

Fig. 9. Computational grid around the NACA0012 airfoil (Grid 1).

Table 2
Summary of the computational grid for the NACA0012 airfoil simulations.

Grid	Minimum cell size $\Delta x/c$	Cell number
1	1.00×10^{-3}	23,938
2	5.00×10^{-4}	55,296
3	2.50×10^{-4}	125,192
4	1.25×10^{-4}	280,924

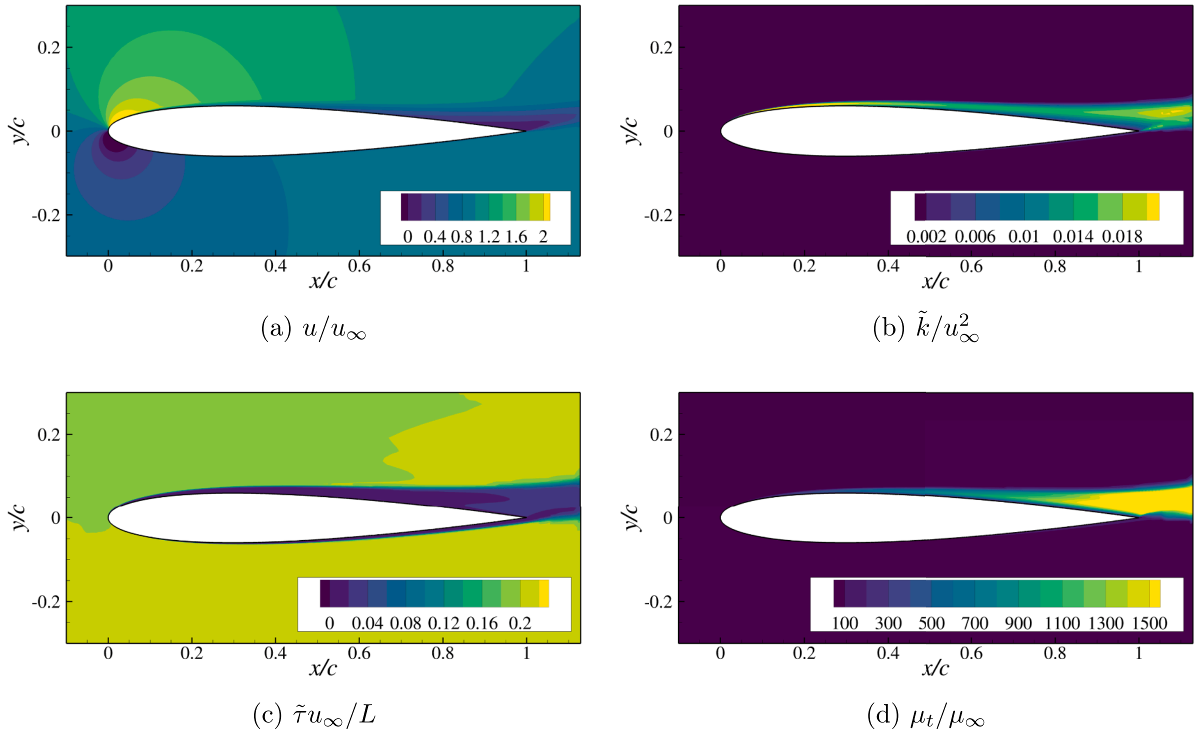


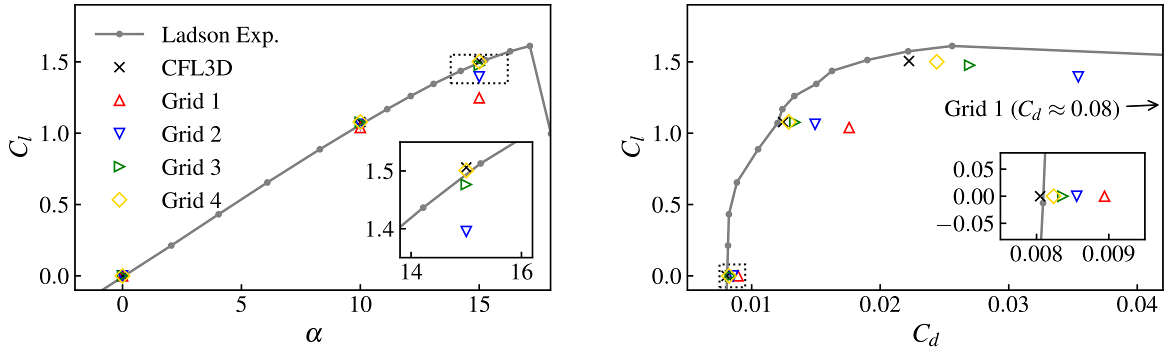
Fig. 10. Distributions of the chordwise velocity and turbulence variables around the NACA0012 airfoil (Grid 4, $\alpha = 15°$).

Table 3
Summary of the aerodynamic coefficients.

	$\alpha = 0°$		$\alpha = 10°$		$\alpha = 15°$	
	C_d	C_l	C_d	C_l	C_d	C_l
UTCart (Grid 4, $\tilde{k} - \tilde{\tau}$ SST model)	0.0082	0.000	0.0129	1.079	0.0244	1.501
CFL3D ($k - \omega$ SST model) [28]	0.0081	0.000	0.0125	1.078	0.0222	1.506
Structured ($\tilde{k} - \tilde{\tau}$ SST model)	0.0081	0.000	0.0128	1.081	0.0241	1.490
Structured ($k - \omega$ SST model)	0.0081	0.000	0.0125	1.082	0.0228	1.502

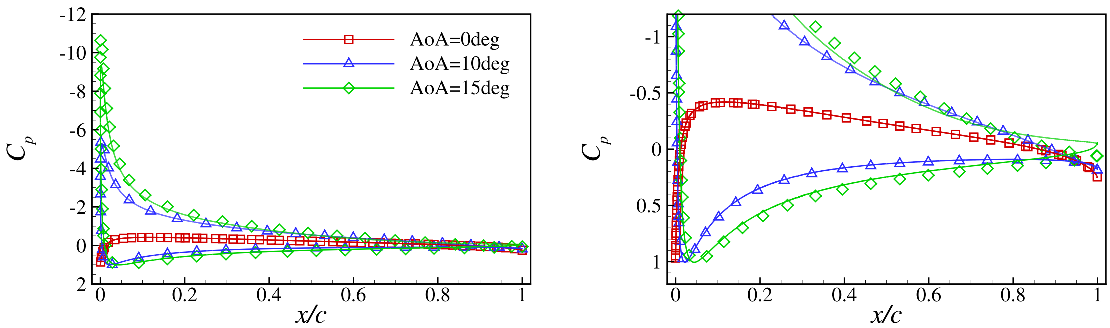
Fig. 11 shows the curves between α, drag coefficient C_d, and lift coefficient C_l obtained by UTCart. In Fig. 11, the experimental results by Ladson [38] and the simulation results obtained by CFL3D [36] with the $k - \omega$ SST model are also shown as reference results. Note that the aerodynamic force in UTCart is evaluated by integrating the numerical fluxes, as described in ref. [32]. The C_d at $\alpha = 0°$ is overpredicted by approximately 10 drag counts (0.001) on Grid 1, whereas the value approaches the experimental value and the reference result by CFL3D as the grid is refined. The overprediction of C_d is primarily due to its pressure component, as evidenced by the C_f distributions not changing significantly across the grids, as described later in the next paragraph. On the coarse grid, C_l tends to be underestimated, and C_d tends to be overestimated at high angles of attack. To examine the differences in results in more detail, the values on Grid 4 and the reference results are summarized in Table 3. At $\alpha = 15°$, the C_d and C_l differences obtained by the UTCart on Grid 4 and CFL3D are 0.0022 and 0.005, respectively. Also, the aerodynamic coefficients obtained by the structured-grid flow solver fall close to the CFL3D results at all the angles of attack, suggesting that the $k - \omega$ SST model and $\tilde{k} - \tilde{\tau}$ SST model behave similarly in this flow and that those models are implemented correctly.

Fig. 12 shows the C_p and C_f distributions over the airfoil on Grids 2 and 4. Here, the results are compared to the reference solution obtained by CFL3D with the $k - \omega$ SST model. The obtained C_p and C_f on Grid 2 at $\alpha = 0°$ and 10° show overall good agreement with the reference results. At $\alpha = 15°$, the C_f on Grid 2 is underestimated at $x/c > 0.2$, whereas the results on Grid 4 show better agreement with the reference results. Moreover, Fig. 13 compares the profiles of the wall-tangential velocity U near the trailing edge (TE) of the airfoil ($x/c = 0.79$ and $x/c = 0.92$) at $\alpha = 10°$ and 15°. Here, because the CFL3D results are not available at TMR, the UTCart results are compared with those obtained by the structured-grid flow solver with the $k - \omega$ SST and $\tilde{k} - \tilde{\tau}$ SST models. At both angles of attack, the profiles obtained by UTCart agree with those by the structured-grid flow solver with the $\tilde{k} - \tilde{\tau}$ SST model. Note that a slight difference remains between the results of the $k - \omega$ SST model and the $\tilde{k} - \tilde{\tau}$ SST model at $\alpha = 15°$. This difference is essentially due to the wall damping functions, but its impact is quite small even under the near-stall condition of $\alpha = 15°$.

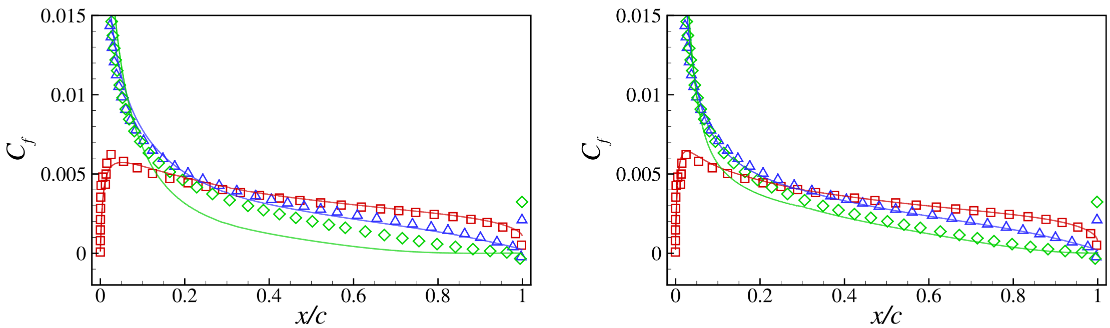


(a) $C_l - \alpha$ (b) $C_d - C_l$

Fig. 11. Aerodynamic coefficients of the NACA0012 airfoil.



(a) C_p (Grid 2) (b) C_p (Grid 2, close-up)

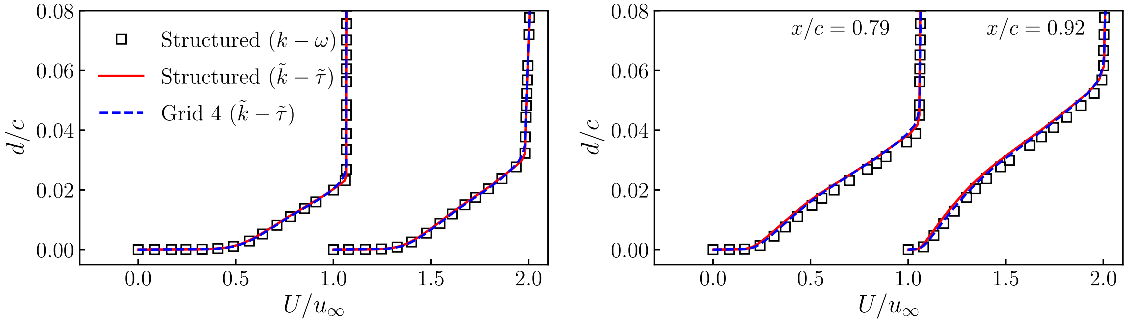


(c) C_f (Grid 2, upper surface) (d) C_f (Grid 4, upper surface)

Fig. 12. Distributions of the surface pressure and skin friction coefficients over the NACA0012 airfoil. Lines: solutions obtained by UTCart; Symbols: reference solution obtained by CFL3D with the $k - \omega$ SST model.

4.4. Flows over a backward-facing step

As a validation problem for separated flows, the flow over a backward-facing step is simulated using UTCart. The inflow boundary is located at $x/H = -130$, where $x = 0$ is the step location, and H is the step height. The lower wall starts at $x/H = -110$ so that the inflow boundary-layer thickness ($\delta/H \approx 2.1$ at $x/H = -4$) is almost the same as the reference experiment [39]. The height of the computational domain upstream of the step is $9H$ and enlarges to $10H$ downstream. The outflow boundary is located at $x/H = 50$, and the back pressure is adjusted such that the Mach number at the reference position $(x/H, y/H) = (-4, 5)$ is approximately 0.128. The normalization in the following uses the x-direction velocity at this position, which is denoted as u_{ref}. The Reynolds number based on H and u_{ref} is 3.6×10^4.



(a) $\alpha = 10°$ (b) $\alpha = 15°$

Fig. 13. Wall-parallel velocity profiles near the TE ($x/c = 0.79$ and $x/c = 0.92$). Each profile is offset horizontally by 1.0.

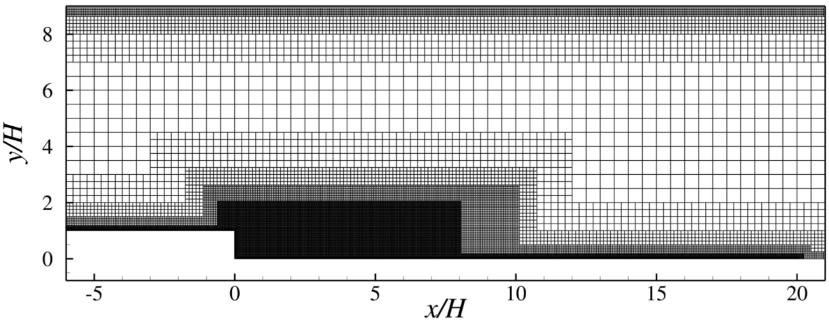


Fig. 14. Computational grid over the backward-facing step (Grid 1).

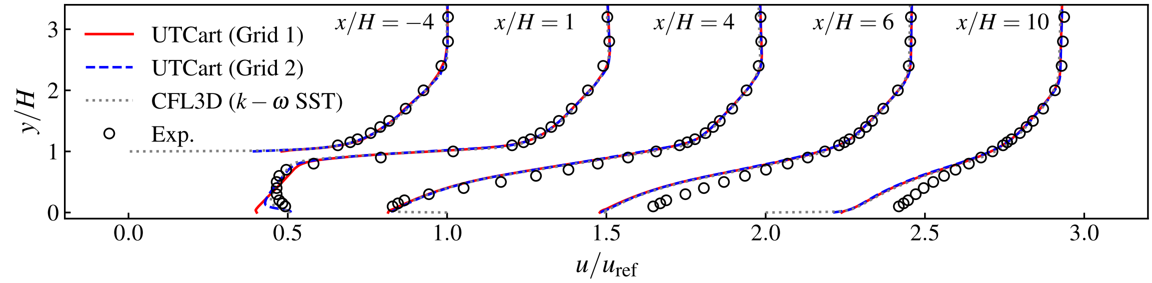
Fig. 14 shows the grid around the step. On Grid 1, the minimum cell size for the lower wall is $H/64$, and a refinement region with $\Delta x/H = 1/32$ is located in $-0.6 < x/H < 8$ and $0 < y/H < 2$. Those cell sizes are halved in Grid 2. For Grids 1 and 2, the total number of cells is 78,776 and 222,665, respectively. For reference, the simulation using the $\tilde{k}-\tilde{\tau}$ SST model on the structured-grid flow solver is also conducted. The computational grid is a body-fitted structured grid provided in TMR, consisting of 177×129 grid points in the step-upstream zone and 257×225 grid points in the step-downstream zone. It has been confirmed that the simulation using this grid and the present structured-grid flow solver with the $k-\omega$ SST model produces results almost identical to the reference CFL3D results.

Fig. 15 compares the velocity and Reynolds shear stress profiles to the reference results obtained by CFL3D with the $k-\omega$ SST model and to the experimental results [39]. Also, although not shown here, the $\tilde{k}-\tilde{\tau}$ SST model results on the structured grid are almost identical to the CFL3D results. At all profiling locations, the results obtained by UTCart show excellent agreement with the CFL3D results. Compared to the experimental results, both the obtained and reference results show a slight discrepancy, especially at the downstream locations ($x/H = 6$ and 10). Since the UTCart and CFL3D results are almost identical, these discrepancies are thought to be due to the $k-\omega$ SST model itself.

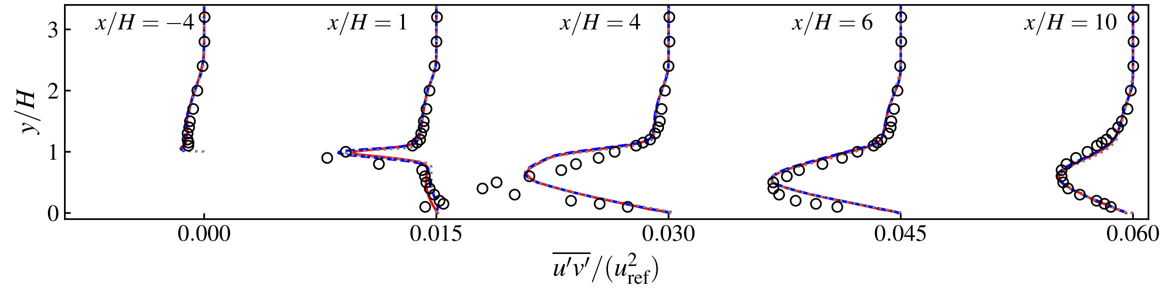
Furthermore, Fig. 16 shows the distributions of C_p and C_f along the bottom wall. For both UTCart and the structured-grid flow solver, C_p obtained by the $\tilde{k}-\tilde{\tau}$ SST model shows good agreement with the CFL3D results. Regarding C_f, the results obtained by the structured-grid flow solver using the $\tilde{k}-\tilde{\tau}$ SST model differ slightly from those by CFL3D. Since the variable transformation from ω to τ gives no differences in the grid-converged state, the minor difference in C_f comes from the wall-damping functions. Although the C_f distribution is not identical, the reattachment occurs almost at the same location as the $k-\omega$ SST model. Here, the C_f obtained by UTCart differs considerably from the CFL3D results in $x/H > 0$. This difference is due to the incompatibility of the employed wall model (Eq. (35)) in the reattaching region. The prediction of C_f may be improved by appropriately considering the non-equilibrium effects in the near-wall region. Nevertheless, the skin friction in this region seems to have little effect on the velocity field, as indicated in the velocity profiles (Fig. 15). Except for the difference in C_f, the results suggest that the $\tilde{k}-\tilde{\tau}$ SST model behaves almost equivalently to the $k-\omega$ SST model also in separated flows.

4.5. Flows over a 3D aircraft

The final test case is the transonic flows around the NASA Common Research Model [40]. The flow conditions are adjusted to the wind tunnel experiment conducted by the Japan Aerospace Exploration Agency [41]. The freestream Mach number is 0.847, and the Reynolds number based on the mean aerodynamic chord c_{ref} and freestream velocity is 2.26×10^6. The simulated angles of attack α are 1.39°, 2.94°, 4.65°, and 5.72°. At each angle of attack, the aerodynamic deformation (i.e., the bending and twisting of

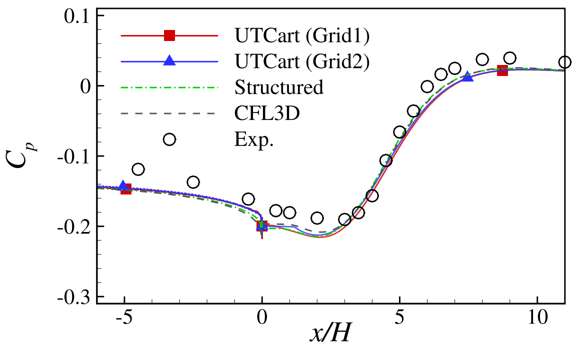


(a) Streamwise velocity (each plot offset by 0.5)

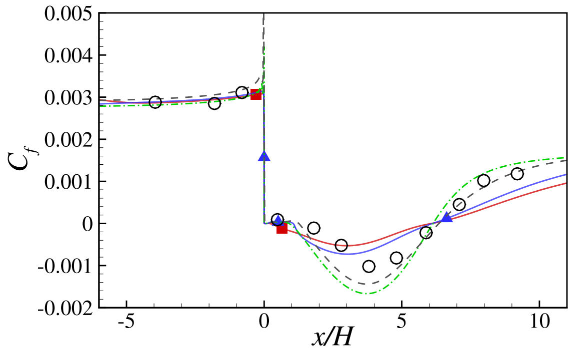


(b) Reynolds shear stress (each plot offset by 0.015)

Fig. 15. Profiles of the streamwise velocity and Reynolds shear stress in the backward-facing step flow.



(a) C_p



(b) C_f

Fig. 16. Distributions of the surface pressure and skin friction coefficients over the lower surface in the backward-facing step flow.

the wing) measured in the experiment is reflected in the simulation geometry. The computational grid is automatically generated by an octree-based algorithm. The computational grid overview is shown in Fig. 17, and the minimum cell sizes and cell counts are summarized in Table 4. Symmetry boundary conditions are applied to the $y = 0$ boundary, and the far-field boundaries exist more than $80c_{\mathrm{ref}}$ away from the model. Note that the cell numbers shown in the table are for $\alpha = 2.94°$, whereas the numbers slightly change with the angles of attack due to the aerodynamic deformation.

When simulating the wing-body geometry, a non-physical flow separation sometimes occurs near the wing-body junction. The previous study [42] reported that the quadratic constitutive relation (QCR) is effective in preventing such side-body separation. Note that the QCR can be combined with any baseline turbulence model based on the Boussinesq approximation since it modifies only the relationship between the Reynolds stress and velocity gradients. Here, we employ the 2024 version of the QCR (QCR2024) [43,44] with the $\tilde{k} - \tilde{\tau}$ SST model, except for the kinetic energy part. QCR2024 improves the representation of anisotropy by introducing two quadratic terms, whereas QCR2000 introduces only one quadratic term. Here, the production terms of the $\tilde{k}$ and $\tilde{\tau}$ equations are retained (i.e., the "-V" variant), although incorporating the QCR effects into the production terms is also possible (see ref. [45]).

Fig. 18 shows the distributions of the x-direction skin friction coefficient $C_{f,x}$ over the upper surface at $\alpha = 2.94°$ and $4.65°$ on Grid 2. At $\alpha = 2.94°$, the flow over the upper surface is mostly attached. At $\alpha = 4.65°$, flow separation is observed downstream of the shock

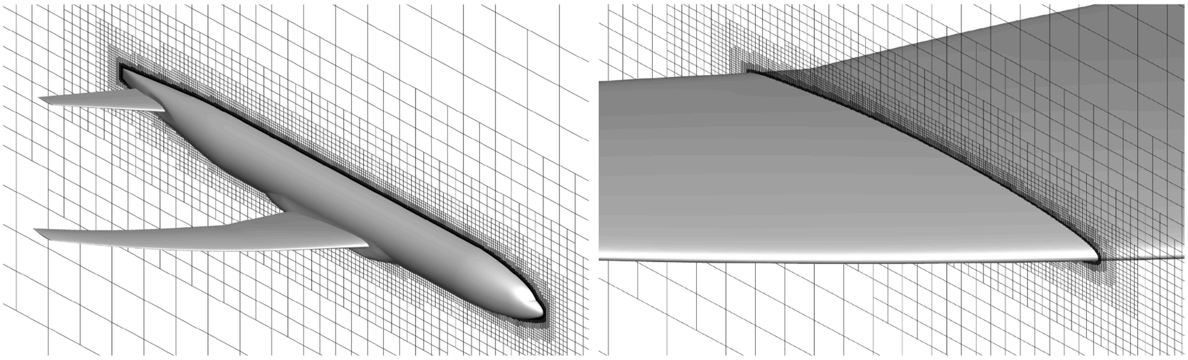
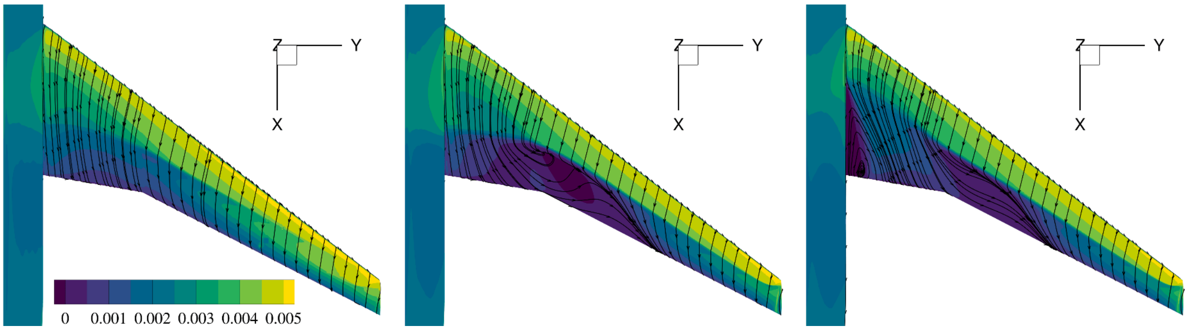


Fig. 17. The computational grid around the NASA Common Research Model (Grid 1).

Table 4
Computational grids around the NASA Common Research Model.

Grid	Wall cell size $\Delta x/c_{\mathrm{ref}}$		Cell number
	Upper wing surface and tail	Lower wing surface and fuselage	
1	1.3×10^{-3}	2.6×10^{-3}	30,986,735
2	1.0×10^{-3}	2.0×10^{-3}	70,684,630



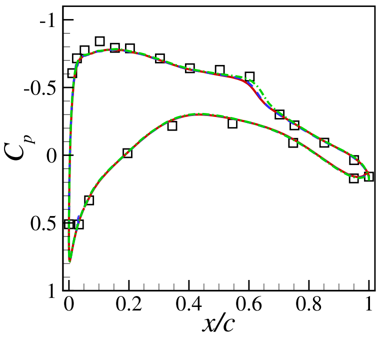
(a) $\alpha = 2.94°$ (with QCR) (b) $\alpha = 4.65°$ (with QCR) (c) $\alpha = 4.65°$ (without QCR)

Fig. 18. Distributions of $C_{f,x}$ and surface streamlines over the upper wing surface (Grid 2).

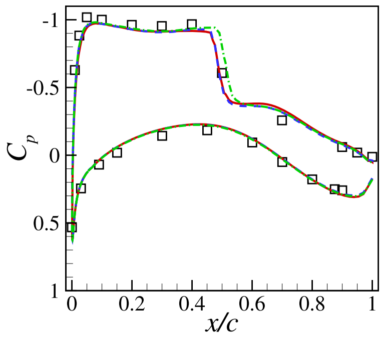
in the mid-span region. Without the QCR, flow separation also occurs in the side-body region. The difference between the results with and without QCR is similar to the previous study [42], which employed the 2000 version of QCR [46] with the SA turbulence model.

Fig. 19 shows the C_p distributions along the wing cross-sections at $\alpha = 2.94°$ and 4.65°. At $\alpha = 2.94°$, the C_p at each cross-section shows good agreement with the experiment, and the result is almost grid-converged. Under this condition, the QCR produces only minor differences in the shock location. At $\alpha = 4.65°$, the simulation using the QCR still shows reasonable agreement with the experiment, except for the shock position and the region behind the shock at Section E. This region involves flow separation caused by the interaction with the shock and is likely sensitive to grid resolution or turbulence modeling. Furthermore, a significant difference is observed in C_p at Section A. Here, the simulation without the QCR shows a different distribution from the experiment, indicating that the local lift is reduced by the side-body separation.

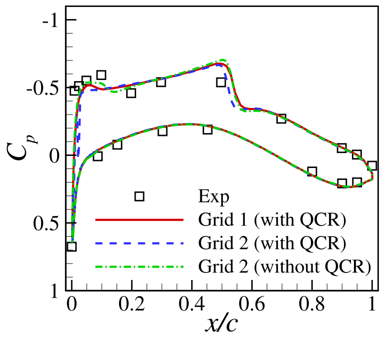
Finally, Fig. 20 shows the lift coefficient C_L, drag coefficient C_D, and pitching moment coefficient C_M. The present results reasonably predict the trend of aerodynamic coefficients, including the change in the lift slope between $\alpha = 2.94°$ and 4.65°. Here, the pitch break between $\alpha = 2.94°$ and 4.65° occurs only with the QCR. This difference is due to the side-body separation, because it reduces the lift force in the region forward of the moment center. Compared to the experiment, C_D tends to be overestimated, and C_M tends to be underestimated. These trends are similar to those observed in a previous study using UTCart with the SA-based turbulence model [32]. The overestimation of C_D is reduced by the grid refinement, where the difference to the experiment at $\alpha = 2.94°$ is approximately 63 and 49 drag counts (0.0063 and 0.0049) on Grid 1 and 2, respectively. The effects of the sting support in the wind tunnel can partly account for the overestimation of C_D and underestimation of C_M. Kohzai et al. [47] conducted the flow simulations with and without the sting support under the same condition ($\alpha \approx 3°$) and reported that the C_D decreases by approximately 0.003 and C_M increases by approximately 0.03 by introducing the sting support. Taking these offsets into account improves the agreement between the simulation and experiment.



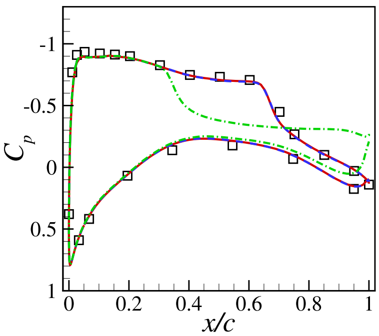
(a) Section A ($\alpha = 2.94°$)



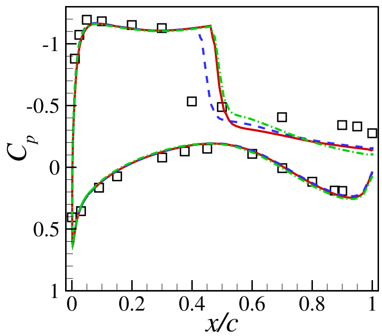
(b) Section E ($\alpha = 2.94°$)



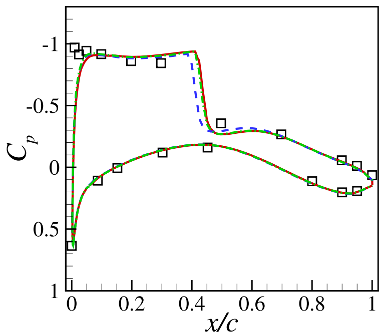
(c) Section I ($\alpha = 2.94°$)



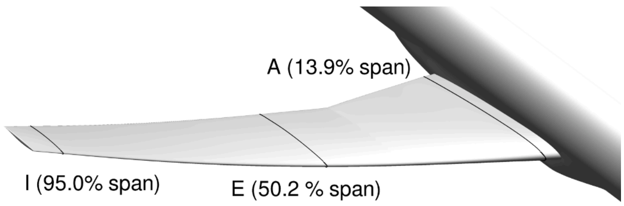
(d) Section A ($\alpha = 4.65°$)



(e) Section E ($\alpha = 4.65°$)

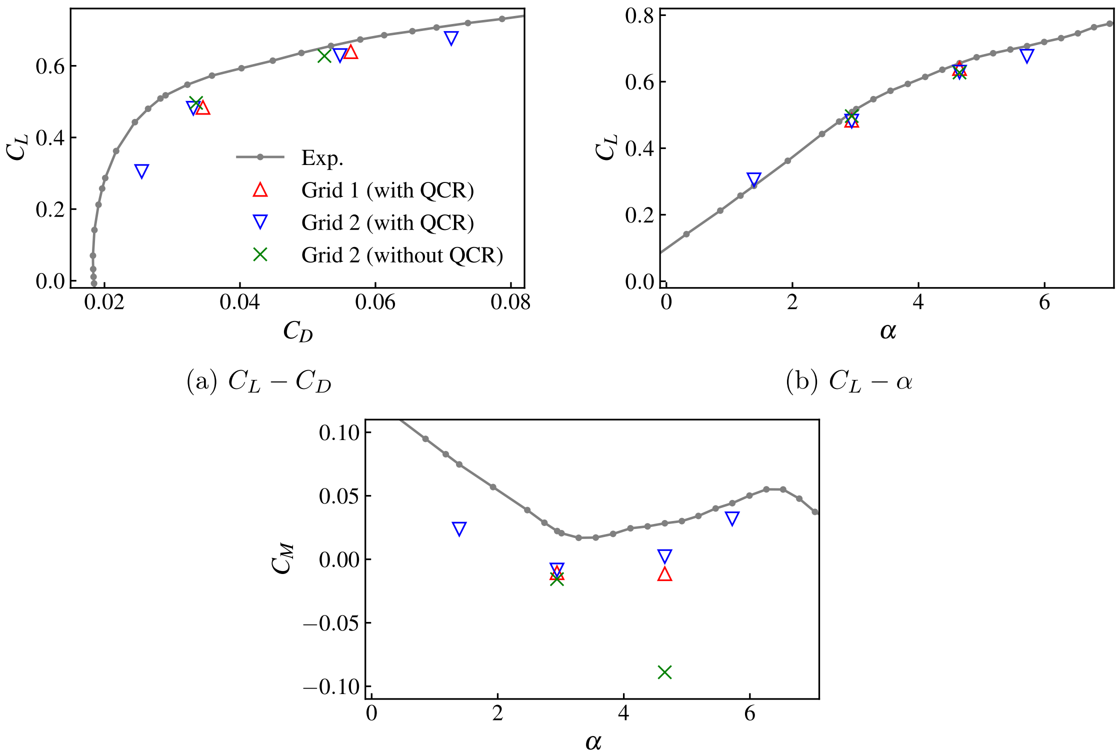


(f) Section I ($\alpha = 4.65°$)



(g) Cross-section definitions

Fig. 19. Surface pressure coefficient distributions over the wing cross-sections.



(a) $C_L - C_D$ (b) $C_L - \alpha$ (c) $C_M - \alpha$

Fig. 20. Aerodynamic coefficients of the NASA Common Research Model.

5. Conclusions

In this study, we presented a numerical strategy for incorporating the $k-\omega$ shear-stress transport (SST) turbulence model in wall-modeled turbulent flow simulations on non-body-conforming Cartesian grids. The present study introduces $\tau = \omega^{-1}$ as an alternative transport variable to ω, avoiding divergence of the variable at the wall. The transport equation of τ was derived analytically from the ω transport equation. Furthermore, to avoid different behaviors of the turbulent transport variables between the viscous sublayer and log layer, a wall damping function f_{v1} was introduced, so that the surrogate transport variables $\tilde{k}$ and $\tilde{\tau}$ have constant or linear solutions throughout the viscous sublayer and log layer. Since those variable profiles can be reproduced even on the coarse grid used in the wall-modeled simulations, the numerical challenges in the near-wall region are significantly reduced. Moreover, the analytical velocity profile in the near-wall region, which can be used as a wall model, was derived from the model equations. For non-body-conforming Cartesian grids, the developed model, named the $\tilde{k}-\tilde{\tau}$ SST model, and its wall model were combined with a partial-slip boundary condition and eddy-viscosity augmentation to maintain the shear-stress balance in the near-wall region.

As the first validation case, the $\tilde{k}-\tilde{\tau}$ SST model was tested on body-conforming grids. The grid-convergence study showed that the skin friction obtained by the $\tilde{k}-\tilde{\tau}$ SST and the original $k-\omega$ SST models converge to almost the same values, whereas the $\tilde{k}-\tilde{\tau}$ SST model exhibits improved grid-convergence behavior. Next, validations were conducted on Cartesian grids using the immersed boundary method with the derived wall function. The validation cases included the flat-plate turbulent boundary layer, the flows around the NACA0012 airfoil, and those over a backward-facing step. In all validation cases, the results showed consistent grid convergence toward the reference body-conforming grid results with the $k-\omega$ SST model. These results suggested that the proposed methodologies, including the $\tilde{k}-\tilde{\tau}$ SST model and wall modeling, faithfully recover the behavior of the original $k-\omega$ SST model even on non-body-conforming Cartesian grids. Furthermore, the $\tilde{k}-\tilde{\tau}$ SST model was combined with the quadratic constitutive relation and applied to simulations of the flows around the NASA Common Research Model. The trends of the aerodynamic coefficients and C_p distributions follow well with the wind tunnel experiment, although there remains room for grid convergence of the drag coefficient. Further improvement in the predicted drag coefficient may be achieved by local grid refinement near the leading edge or by improving the interpolation formula of the flow quantities (e.g., pressure) at the wall boundary. Nevertheless, considering that Cartesian grids can be generated automatically, the present methodologies could be a promising tool for aerodynamic aircraft design with reasonable accuracy.

CRediT authorship contribution statement

Yoshiharu Tamaki: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Christophe Friess:** Writing – review & editing, Conceptualization; **Jérôme Jacob:** Writing – review & editing, Conceptualization; **Taro Imamura:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Yoshiharu Tamaki reports financial support was provided by Japan Society for the Promotion of Science. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A. Discretization of the diffusion terms

Since the diffusion term in the $\tilde{\tau}$ equation is different from the usual conservative form, its discretization requires special care. A straightforward method is to divide the term into the conservative and the other parts as

$$\begin{aligned}&\tilde{\tau}^2 \frac{\partial}{\partial x_j}\left[\left(\mu(1-F_1)+\sigma_\omega \rho \tilde{k}\tilde{\tau}\right)\frac{1}{\tilde{\tau}^2}\frac{\partial \tilde{\tau}}{\partial x_j}\right]\\ &= \frac{\partial}{\partial x_j}\left[\left(\mu(1-F_1)+\sigma_\omega \rho \tilde{k}\tilde{\tau}\right)\frac{\partial \tilde{\tau}}{\partial x_j}\right] - \frac{2}{\tilde{\tau}}\left(\mu(1-F_1)+\sigma_\omega \rho \tilde{k}\tilde{\tau}\right)\frac{\partial \tilde{\tau}}{\partial x_j}\frac{\partial \tilde{\tau}}{\partial x_j}.\end{aligned} \tag{A.1}$$

Here, the second term on the right-hand side is introduced as a source term. However, when this source term is evaluated using the cell-centered derivatives, a spurious peak of $\tilde{\tau}$ appears at the boundary layer edge. Therefore, we employ another method based on non-conservative fluxes. When computing the diffusion between the left (L) and right (R) cells, the left and right fluxes at the middle face (LR) are

$$f_L^{\tilde{\tau},\mathrm{diff}} = \tilde{\tau}_L^2 \tilde{f}_{LR},\ f_R^{\tilde{\tau},\mathrm{diff}} = \tilde{\tau}_R^2 \tilde{f}_{LR} \tag{A.2}$$

where

$$\tilde{f}_{LR} = \left[(\mu(1-F_1)+\sigma_\omega \rho \tilde{k}\tilde{\tau})\frac{1}{\tilde{\tau}^2}\frac{\partial \tilde{\tau}}{\partial x_j}\right]_{LR}. \tag{A.3}$$

Here, the gradient at LR should correctly reflect the difference between $\tilde{\tau}_L$ and $\tilde{\tau}_R$ to avoid even-odd decoupling. For example, the method described in ref. [48] is used in UTCart. By using this method, the coefficient $\tilde{\tau}_L^2$ or $\tilde{\tau}_R^2$ is not differentiated at each cell center L or R, recovering the original formulation.

Besides, since $\tilde{f}_{LR} \propto d^{-1}$ near the wall, the computation may become inaccurate near the wall. Therefore, the diffusion term is transformed using a wall damping function ϕ as

$$\begin{aligned}&\tilde{\tau}^2 \frac{\partial}{\partial x_j}\left[\left(\mu(1-F_1)+\sigma_\omega \rho \tilde{k}\tilde{\tau}\right)\frac{1}{\tilde{\tau}^2}\frac{\partial \tilde{\tau}}{\partial x_j}\right]\\ &= \frac{\tilde{\tau}^2}{\phi}\frac{\partial}{\partial x_j}\left[\left(\mu(1-F_1)+\sigma_\omega \rho \tilde{k}\tilde{\tau}\right)\frac{\phi}{\tilde{\tau}^2}\frac{\partial \tilde{\tau}}{\partial x_j}\right] - \frac{1}{\phi}\left(\mu(1-F_1)+\sigma_\omega \rho \tilde{k}\tilde{\tau}\right)\frac{\partial \tilde{\tau}}{\partial x_j}\frac{\partial \phi}{\partial x_j}.\end{aligned} \tag{A.4}$$

Here, the wall damping function should be proportional to d in the near-wall region. Thus, we employ

$$\phi = \sin\left(\frac{\pi \min(d, d_{\mathrm{cutoff}})}{2d_{\mathrm{cutoff}}}\right). \tag{A.5}$$

In this study, d_{cutoff} is set to be three times the minimum cell size (i.e., equal to the IP distance in UTCart). The corresponding left and right fluxes for the first term on the right-hand side of Eq. (A.4) are

$$f_L^{\tilde{\tau},\text{diff1}} = \frac{\phi_{LR}}{\phi_L}\tilde{\tau}_L^2 \tilde{f}_{LR},\ f_R^{\tilde{\tau},\text{diff1}} = \frac{\phi_{LR}}{\phi_R}\tilde{\tau}_R^2 \tilde{f}_{LR}. \tag{A.6}$$

Note that the following relation should be used to avoid division by zero in case of the face with a zero or tiny wall distance (i.e., where $\tilde{\tau}_{LR} \approx 0$):

$$\frac{\phi_{LR}}{\tilde{\tau}_{LR}} \approx \frac{\phi_L}{\tilde{\tau}_L}. \tag{A.7}$$

Also, the second term in Eq. (A.4) is evaluated with assuming $F_1 = 1$ as

$$-\frac{1}{\phi}(\mu(1-F_1)+\sigma_\omega \rho\tilde{k}\tilde{\tau})\frac{\partial\tilde{\tau}}{\partial x_j}\frac{\partial\phi}{\partial x_j} = -\frac{\sigma_{\omega 1}\rho\tilde{k}\tilde{\tau}}{\phi}\left[\frac{\partial}{\partial x_j}\left(\phi\frac{\partial\tau}{\partial x_j}\right) - \phi\frac{\partial}{\partial x_j}\left(\frac{\partial\tau}{\partial x_j}\right)\right], \tag{A.8}$$

for which the fluxes are

$$\begin{aligned} f_L^{\tilde{\tau},\text{diff2}} &= -\sigma_{\omega 1}\left[\frac{\rho\tilde{k}\tilde{\tau}}{\phi}\right]_L \left.\frac{\partial\tilde{\tau}}{\partial x_j}\right|_{LR}(\phi_{LR}-\phi_L),\\ f_R^{\tilde{\tau},\text{diff2}} &= -\sigma_{\omega 1}\left[\frac{\rho\tilde{k}\tilde{\tau}}{\phi}\right]_R \left.\frac{\partial\tilde{\tau}}{\partial x_j}\right|_{LR}(\phi_{LR}-\phi_R). \end{aligned} \tag{A.9}$$

Here $F_1 = 1$ is assumed because ϕ is active only in the near-wall region.

Similarly, the cross-diffusion term can be equivalently transformed into

$$(1-F_1)\text{CD}_{\tilde{k}\tilde{\tau}} = 2(1-F_1)\sigma_{\omega 2}\rho\tilde{\tau}\left[\frac{\partial}{\partial x_j}\left(\tilde{k}\frac{\partial\tilde{\tau}}{\partial x_j}\right) - \tilde{k}\frac{\partial}{\partial x_j}\left(\frac{\partial\tilde{\tau}}{\partial x_j}\right)\right]. \tag{A.10}$$

Based on this form, the left and right fluxes are written as

$$\begin{aligned} f_L^{\tilde{\tau},\text{CD}} &= \left[2(1-F_1)\sigma_{\omega 2}\rho\tilde{\tau}\right]_L \left.\frac{\partial\tilde{\tau}}{\partial x_j}\right|_{LR}(\tilde{k}_{LR}-\tilde{k}_L),\\ f_R^{\tilde{\tau},\text{CD}} &= \left[2(1-F_1)\sigma_{\omega 2}\rho\tilde{\tau}\right]_R \left.\frac{\partial\tilde{\tau}}{\partial x_j}\right|_{LR}(\tilde{k}_{LR}-\tilde{k}_R). \end{aligned} \tag{A.11}$$

The final fluxes are the sum of Eqs. (A.6), (A.9), and (A.11). These fluxes share the face-center (LR) quantities, and therefore, the computational cost does not increase significantly compared to that for an ordinary diffusion flux.

References

[1] R. Mittal, G. Iaccarino, Immersed boundary methods, Annu. Rev. Fluid Mech. 37 (1) (2005) 239–261.
[2] R. Verzicco, Immersed boundary methods: historical perspective and future outlook, Annu. Rev. Fluid Mech. 55 (1) (2023) 129–155.
[3] G. Kalitzin, G. Iaccarino, Turbulence modeling in an immersed-boundary RANS method, CTR Annu. Br. (2002) 415–426.
[4] F. Tessicini, G. Iaccarino, M. Fatica, M. Wang, R. Verzicco, Wall modeling for large-eddy simulation using an immersed boundary method, Cent. Turbul. Res. Annu. Res. Br. 2002 (2002) 181–187.
[5] F. Roman, V. Armenio, J. Fröhlich, A simple wall-layer model for large eddy simulation with immersed boundary method, Phys. Fluid. 21 (10) (2009).
[6] M. Berger, M. Aftosmis, Progress towards a Cartesian cut-cell method for viscous compressible flow, in: 50th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition, 2012. AIAA Paper No. 2012-1301.
[7] M. Harada, Y. Tamaki, Y. Takahashi, T. Imamura, Simple and robust cut-cell method for high-Reynolds-number-flow simulation on Cartesian grids, AIAA J. 55 (8) (2017) 2833–2841.
[8] Y. Tamaki, M. Harada, T. Imamura, Near-wall modification of Spalart–Allmaras turbulence model for immersed boundary method, AIAA J. 55 (9) (2017) 3027–3039.
[9] B. Constant, S. Péron, H. Beaugendre, C. Benoit, An improved immersed boundary method for turbulent flow simulations on Cartesian grids, J. Comput. Phys. 435 (2021) 110240.
[10] S. Mozaffari, J. Jacob, P. Sagaut, Assessment of wall modeling with adverse pressure gradient for high Reynolds number separated flows, Flow Turbul. Combust. 113 (4) (2024) 923–945.
[11] F. Capizzano, Turbulent wall model for immersed boundary methods, AIAA J. 49 (11) (2011) 2367–2381.
[12] Y. Tamaki, S. Kawai, Wall modeling for large-eddy simulation on non-body-conforming Cartesian grids, Phys. Rev. Fluid. 6 (11) (2021) 114603.
[13] S.-G. Cai, J. Degrigny, J.-F. Boussuge, P. Sagaut, Coupling of turbulence wall models and immersed boundaries on Cartesian grids, J. Comput. Phys. 429 (2021) 109995.
[14] M. Terracol, L. Manueco, A dynamic linearized wall model for turbulent flow simulation: towards grid convergence in wall-modeled simulations, J. Comput. Phys. 521 (2025) 113555.
[15] P. Spalart, S. Allmaras, A one-equation turbulence model for aerodynamic flows, in: 30th Aerospace Sciences Meeting and Exhibit, 1992. AIAA Paper No. 92–439.
[16] G. Kalitzin, G. Medic, G. Iaccarino, P. Durbin, Near-wall behavior of RANS turbulence models and implications for wall functions, J. Comput. Phys. 204 (1) (2005) 265–291.
[17] S.R. Allmaras, F.T. Johnson, Modifications and clarifications for the implementation of the Spalart-Allmaras turbulence model, in: Seventh International Conference on Computational Fluid Dynamics (ICCFD7), 2012. ICCFD Paper No. 1902.
[18] F.R. Menter, Improved Two-Equation K-Omega Turbulence Models for Aerodynamic Flows, Technical Report, National Aeronautics and Space Administration, 1992. NASA Technical Memorandum 103975.

[19] D.C. Wilcox, Formulation of the kw turbulence model revisited, AIAA J. 46 (11) (2008) 2823–2838.
[20] B. Eisfeld, C.L. Rumsey, V. Togiti, S. Braun, A. Stürmer, Reynolds-stress model computations of NASA juncture flow experiment, AIAA J. 60 (3) (2022) 1643–1662.
[21] F.R. Menter, Two-equation eddy-viscosity turbulence models for engineering applications, AIAA J. 32 (8) (1994) 1598–1605.
[22] F.R. Menter, M. Kuntz, R. Langtry, et al., Ten years of industrial experience with the SST turbulence model, Turbul. Heat Mass Transf. 4 (1) (2003) 625–632.
[23] M.S. Gritskevich, A.V. Garbaruk, J. Schütze, F.R. Menter, Development of DDES and IDDES formulations for the k-ω shear stress transport model, Flow Turbul. Combust. 88 (3) (2012) 431–449.
[24] S. Deck, N. Renard, Towards an enhanced protection of attached boundary layers in hybrid RANS/LES methods, J. Comput. Phys. 400 (2020) 108970.
[25] N. Renard, J. Vaquero, F. Gand, S. Deck, A fully automatic and robust hybrid Reynolds-averaged Navier–Stokes/large eddy simulation approach based on the Menter shear stress transport k–ω model, Phys. Fluid. 36 (11) (2024). https://doi.org/10.1063/5.0222762
[26] B. de Laage de Meux, B. Audebert, R. Manceau, R. Perrin, Anisotropic linear forcing for synthetic turbulence generation in large eddy simulation and hybrid RANS/LES modeling, Phys. Fluid. 27 (3) (2015). https://doi.org/10.1063/1.4916019
[27] A. Monot, C. Friess, J. Wackers, Grey area mitigation for Detached-Eddy simulations using volume forcing, Int. J. Comput. Fluid Dyn. 38 (2–3) (2024) 246–264.
[28] Tubulence modeling resource for the validation and verification of turbulence models, 2026. (https://tmbwg.github.io/turbmodels). Retrievied on June 10th, 2026.
[29] C.G. Speziale, R. Abid, E.C. Anderson, Critical evaluation of two-equation models for near-wall turbulence, AIAA J. 30 (2) (1992) 324–331.
[30] P.R. Spalart, C.L. Rumsey, Effective inflow conditions for turbulence models in aerodynamic calculations, AIAA J. 45 (10) (2007) 2544–2553.
[31] D.B. Spalding, et al., A single formula for the law of the wall, J. Appl. Mech. 28 (3) (1961) 455–458.
[32] Y. Tamaki, T. Imamura, Turbulent flow simulations of the common research model using immersed boundary method, AIAA J. 56 (6) (2018) 2271–2282.
[33] Y. Tamaki, T. Imamura, Numerical evaluation of installation effects in semispan model wind-tunnel testing, AIAA J. 64 (1) (2026) 13–25.
[34] E. Shima, K. Kitamura, Parameter-free simple low-dissipation AUSM-family scheme for all speeds, AIAA J. 49 (8) (2011) 1693–1709.
[35] B. van Leer, Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov's method, J. Comput. Phys. 32 (1) (1979) 101–136.
[36] CFL3D Version 6 Home page, 2025. (https://nasa.github.io/CFL3D/). Retrievied on June 12th, 2026.
[37] FUN3D Manual, 2025. (https://fun3d.larc.nasa.gov). Retrievied on June 12th, 2026.
[38] C.L. Ladson, Effects of Independent Variation of Mach and Reynolds Numbers on the Low-Speed Aerodynamic Characteristics of the NACA 0012 Airfoil Section, Technical Report, National Aeronautics and Space Administration, 1988. NASA TM 4074.
[39] D.M. Driver, H.L. Seegmiller, Features of a reattaching turbulent shear layer in divergent channelflow, AIAA J. 23 (2) (1985) 163–171.
[40] J. Vassberg, M. Dehaan, M. Rivers, R. Wahls, Development of a common research model for applied CFD validation studies, in: 26th AIAA Applied Aerodynamics Conference, 2008. AIAA Paper No. 2008–6919.
[41] M. Ueno, M. Kohzai, S. Koga, JAXA Transonic Wind Tunnel Test of the NASA CRM (Volume 1), Technical Report, Japan Aerospace Exploration Agency, 2014. Research Memorandum JAXA-RM-13-017E, Tokyo.
[42] K. Yamamoto, K. Tanaka, M. Murayama, Effect of a nonlinear constitutive relation for turbulence modeling on predicting flow separation at wing-body juncture of transonic commercial aircraft, in: 30th AIAA Applied Aerodynamics Conference, 2012. AIAA Paper No. 2012–2895.
[43] Y. Tamaki, S. Kawai, Turbulence anisotropy effects on corner-flow separation: physics and turbulence modelling, J. Fluid Mech. 980 (2024) A21.
[44] Y. Tamaki, S. Kawai, T. Imamura, Applicability of quadratic constitutive relations for turbulent flow simulations in aeronautics, J. Aircr. 63 (2) (2026) 749–762.
[45] Y. Tamaki, T. Imamura, NASA juncture flow computation using turbulence models with quadratic constitutive relations, in: AIAA AVIATION Forum and ASCEND 2025, 2025. AIAA Paper No. 2025–3110.
[46] P.R. Spalart, Strategies for turbulence modelling and simulations, Int. J. Heat Fluid Flow 21 (3) (2000) 252–263.
[47] M. Kohzai, M. Ueno, S. Koga, N. Sudani, Wall and Support Interference Corrections of NASA Common Research Model Wind Tunnel Tests in JAXA, 2013. AIAA Paper No. 2013-0963.
[48] G. Wang, A. Schwöppe, R. Heinrich, Comparison and evaluation of cell-centered and cell-vertex discretization in the unstructured tau-code for turbulent viscous flows, in: ECCOMAS CFD, 2010.